University College London

Regime Detection using Sliced Wasserstein k-means Clustering with UnifOrtho

A thesis submitted in partial fulfillment of the
MSc in Computational Finance

Oscar Peyron

Supervisor:
Prof. Philip Treleaven (University College London)

University College London

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Abstract

This thesis uses a modified k-means clustering algorithm to detect regimes in financial time-series data. As practitioners increasingly turn to high-dimensional data to build multi-asset strategies, detecting regimes in high dimensions has become a key challenge. Moreover, there is a growing need for methods that are not only accurate but also transparent enough to be trusted. The motivation of this research, is therefore to investigate and improve regime detection algorithms in high dimensions that are both interpretable and accurate.

To this end, the thesis clusters multi-dimensional returns distributions by building on the Wasserstein k-means algorithm of Horvath et al. [10] and its high-dimensional generalization, the sliced Wasserstein clustering algorithm of Luan and Hamp (2025) [11]. Those models infer the current market regime without pre-specified assumptions on the returns distribution. We extend this framework by improving the sliced Wasserstein distance approximation through the UnifOrtho method of Rowland et al. [16]. Rather than relying on a fixed set of well-chosen random projections, as in Luan and Hamp (2025) [11], we show that UnifOrtho yields more accurate regime detection in high dimensions.

The thesis, conducted in collaboration with a major sovereign wealth fund and its research institute, consists of the following methodology chapter and three research experiments:

- **Methodology and Theoretical Background.** In this chapter, we will introduce the mathematical background needed to understand the sliced Wasserstein k-means clustering algorithm and its UnifOrtho improvement. This section also introduces the relevant framework for the implied probability of regime switches.

- **Experiment 1: Synthetic Data Testing.** The first experiment is a supervised-learning setup similar to the one in Luan and Hamp (2025) [11]. This includes producing synthetic data with multiple known regimes (from 2 regimes to 3 regimes) and comparing the clustering results with ground truth to assess accuracy.

- **Experiment 2: Sliced Wasserstein on real-world data sets.** The second experiment extends the study to real-world data sets and in particular on equity index data from three different regional markets (US, Europe and Asia). First, look-ahead trading strategies based on the produced labelling of the sliced Wasserstein k-means algorithms will be used to assess the accuracy of the clustering results. Then, the focus will shift to non-look-ahead trading strategies to assess the performance of the clustering results in a realistic setting.

- **Experiment 3: Interpretability of clustering results.** Finally, the third study will focus on training supervised learning algorithms using clusterings results from the

sliced Wasserstein k-means to produce a feature importance analysis using the SHAP framework.

In this research we therefore provide the following contributions to science:

- **Scalable regime detection algorithm.** The main objective of this thesis is to provide a regime detection algorithm that scales to very high-dimensional settings without relying on manually pre-constructed projection vectors. Instead, UnifOrtho automatically produces a large number of orthogonal projection vectors as the dimensionality grows.
- **Computationally tractable regime detection algorithm.** We lay out the mathematical foundations for the sliced Wasserstein distance and its enhancement, with a time complexity that is polynomial in the dimension and the input size.

We also provide the following contributions to industry:

- **Generation of predictive signals from clustering results.** The thesis shows how to construct implied probabilities of regime switches from the clustering results of the sliced Wasserstein algorithm, turning the detected regimes into tradeable signals.
- **Interpretability of clustering results.** We show how the statistical results of the sliced Wasserstein k-means clustering algorithm align with empirical observations of financial markets. The mathematical framework behind the algorithm is highly transparent and intelligible while still delivering accurate results, meeting the growing need for models that practitioners can trust.

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Chapter 1

Introduction

This thesis investigates regime detection in high-dimensional financial time-series through an interpretable and computationally efficient machine learning algorithm. The research was conducted in collaboration with a major sovereign wealth fund and its research institute with key industry perspectives that shaped the practical direction of the work. This chapter motivates the problem, states the research objectives, outlines the experiments and their contribution to the literature, and closes with the structure of the thesis.

1.1 Motivations

Regime detection is the study of the changes in the underlying dynamics of a financial time-series. This encompasses the study of shifts in mean, volatility, correlation structure and other statistical properties of time-series data. The challenge in detecting regimes is the capacity to distinguish macroscopic regimes (i.e bear/bull regimes or high/low volatility regimes) from microscopic ones such as intraday mean-reversion or trends.

Because financial time-series are notoriously non-stationary many static quantitative models fail to capture dynamics when faced to the real-world. This is why regime detection is such of great value for practitioners. On the buy side, funds and asset managers rely on diversification to hedge their clients against risk. However, this diversification relies on a stable correlation between assets, which is not the case during periods of crisis. One often observes that during extreme bear regimes cross-asset correlations converge towards 1, which constitutes a major issue since the diversifiable risk is not diversified anymore. This is why regime detection is used as a risk management filter. For instance, switch periods from low/vol regimes to high/vol regimes are flagged to change portfolio allocation parameters. On the sell side, firms must handle large amounts of client orders and need to adapt their trading strategies. This is for instance, the case for market making execution logic which alters its

VWAP/TWAP algorithms to adapt to micro-regimes. At an intraday level, the market can change from a mean-reverting regime to a trending regime and vice versa. Adapting to these macroscopic and microscopic regimes is crucial for practitioners to avoid losses and improve their performance.

With the emergence of electronic trading and the availability of low to high-frequency data, it has become more accessible to train machine learning models on financial time-series data. This has led to a growing interest in using machine learning models to detect regimes in financial markets. In the meantime, practitioners have showed growing interest over models that not only work but also that are intelligible. Moreover, with the emergence of portfolio trading – a trading strategy that trades a basket of up to 100 assets at the same time – it has become crucial to have regime detection methods that can handle high-dimensional time-series data. This additional component makes therefore interpretability and scalability crucial.

The principal motivation of this thesis is to investigate and improve regime detection algorithms in high-dimensional settings that are simultaneously interpretable and accurate. First, it improves an existing high-dimensional regime detection framework, addressing its scalability to the higher dimensions, real-world applications demand. It also adds forecasting power via statistical learning, letting practitioners adapt trading strategies before market dynamics shift, not after.

Finally, there are limited number of methods that provide both interpretability and model-free approaches as the one proposed by this thesis. Hidden-Markov Models (HMM) are one of the most widely used methods for regime detection, yet their impose a Gaussian-mixture assumptions on returns as well as require an re-estimation of parameters at breaking points [3]. On the other hand, deep learning methods are well-known to be opaque [8] and understanding their decision-making process is a key research topic in mathematics.

1.2 Research Objectives

The main objectives of this thesis is to investigate and improve the existing sliced Wasserstein k-means clustering algorithm for regime detection in high-dimension by Luan et. al (2025) [11]. The latter algorithm is a generalization of the Wasserstein k-means clustering algorithm proposed by Horvath et. al (2024) [10] to high-dimensional time-series data. Both of those algorithm are based on the Wasserstein distance and its extension to the sliced Wasserstein distance. This high-dimensional distance requires a Monte Carlo approximation via orthogonal projections vectors to be computed. Instead of using a set of pre-fixed projections vectors as in Luan et. al [11], this thesis uses the UnifOrtho method proposed by Rowland et. al [16] to sample orthogonal projections vectors for the Monte Carlo approximation of the sliced Wasserstein distance. This UnifOrtho approach for computing the sliced Wasserstein distance can be more easily scaled as the dimension increases and is more robust to the curse of dimensionality. This thesis also aims to quantify the enhancement of the UnifOrtho approach on synthetic data as well as on real-world data sets. Another key objective is to provide predictive power on future regimes using the mathematical framework of the sliced Wasserstein k-means algorithm. The aim is to generate trading strategies based on the ability to detect future market downturns or upturns. Finally, this research further explores the interpretability of the improved method by comparing the clustering results with macroeconomic indicators. The underlying hypothesis that this thesis seek to demonstrate is that the statistical results given by the sliced Wasserstein k-means clustering algorithm align with empirical observations of macroeconomic dynamics.

1.3 Research Experiments

This dissertation starts from exploring the improved accuracy of the sliced Wasserstein k-means clustering algorithm using the UnifOrtho approach on synthetic data in the first experiment. The second experiment extends this study on real-world data sets. The last experiment concludes with an interpretability test by combining previous results with macroeconomic indicators.

Before diving in more detail into the experiments, this thesis provides a theoretical methodology chapter, in order to expose the mathematical backgrounds that explains the novel regime detection framework of the Wasserstein k-means clustering algorithm by Horvath et .al [10] and Luan et .al in high-dimensions [11]. It also gives insight on the UnifOrtho approach by Rowland et. al [16] and how it can be used to improve the sliced Wasserstein distance approximation in high-dimensions. More importantly, it provides the mathematical rationale behind using the UnifOrtho approach before testing it empirically. The following empirical study is split into three experiments:

- **Synthetic Data Testing.** The first experiment is aimed at better understanding the total accuracy of the sliced Wasserstein k-means clustering variant with UnifOrtho on synthetic data. The aim is to compare the results with the same experiment done in Luan et.al [11] and show that the UnifOrtho approach yields better results for increasingly higher dimensional data. One produces different regimes in the synthetic data, by changing the mean and/or the correlation structure of the multivariate data. This research also looks at synthetic data which combines gaussian regimes with non-gaussian regimes. Finally, testing is done on data with different dimensionality (i.e from 2 to 100).

- **Sliced Wasserstein on real world data sets.** After testing on synthetic data, the next step is to apply the sliced Wasserstein k-means clustering algorithm with UnifOrtho on real-world data sets. In particular, the model is applied to three regional equity index data sets (US, Europe and Asia) from 1990 to 2020 to identify potential gaps in performance between geographic locations. First, a look-ahead strategy is applied using the clustering results. This allows to get a quantifiable performance of the sliced Wasserstein method on the above-mentioned data sets. The focus then shifts to non-look-ahead trading strategies to assess the performance of the clustering results in a more realistic setting with predictive signals generated from implied probabilities of regime switches. This also extends to tuning the hyperparameters of the developed trading strategies and assessing the performance of the tuning on test-data sets (2020-2025 period).

- **Interpretability of clustering results.** The final experiment will consist on fitting supervised learning algorithms – namely OLS Linear Regression, Logistic Regression and Random Forest – to the clustering results of the sliced Wasserstein k-means clustering algorithm with a set of 10 macroeconomic indicators on the study period from 2006 to 2020. The objective is to identify using the SHAP feature importance framework which macroeconomic indicators are highly sensitive with respect to the model during bear clusters.

1.4 Scientific and Industrial Contributions

This dissertation provides a number of contributed to the scientific literature and to industry in the following areas:

- **Scalable regime detection algorithm.** This thesis provides a regime-detection algorithm that can be scaled to very high-dimensions without hand-constructed projections. The objective is eventually to use the sliced Wasserstein k-means on high-dimensional data that requires hundreds of projections vectors for good accuracy. In particular, this sheds light on financial applications of the UnifOrtho method proposed by Rowland et. al [16] to improve the Monte Carlo approximation of the sliced Wasserstein distance in high-dimensions.

- **Computationally tractable framework.** Additionally, the aim of this research is to replace the manual hand-crafted projections vectors used in Luan et. al [11] with polynomial-time complexity algorithms such as the UnifOrtho method which runs in $\mathcal{O}(d^3)$ time complexity with d representing the dimension of the time series data.

- **Predictive signal generation from clustering.** The thesis shows how to construct trading signals from the clustering algorithms by deriving implied probabilities of regime switches. This a key contribution to the research by Luan et al. [11], which already hinted towards such use of the sliced Wasserstein framework as possible future work. Additionally, the thesis provides multiple trading strategies that exploit this predictive signal to generate returns on real-world data sets.

- **Interpretability of clustering results.** Finally, the thesis show how the statistical results given by the sliced Wasserstein k-means clustering algorithm empirically match with macroeconomic indicators. This provides not only a model that is mathematically interpretable but also a model that follows empirical macroeconomic rationale.

1.5 Thesis Structure

- **Background and Literature Review.** This section focuses on laying out the key literature and research on regime detection. It also provides an overall context around the sliced Wasserstein k-means clustering algorithm within the existing literature.

- **Methodology and Theoretical Background.** This chapter of the thesis provides the theoretical framework for understanding the Wasserstein k-means algorithm and its high-dimensional generalization to the sliced Wasserstein k-means algorithm. In particular, it explains why the UnifOrtho method approach for the sliced Wasserstein algorithm yields theoretical improvements with respect to existing methods in the literature. It further describes in detail the mathematical framework behind the implied probability of regime switches which will be used in the trading strategies later.

- **Experiment 1: Synthetic Data Testing.** This section provides a synthetic data testing of the sliced Wasserstein model with key accuracy comparisons for different tunings. This synthetic experiment also includes an accuracy comparison with existing results in the literature (i.e Luan et al. (2025) [11]).

- **Experiment 2: Sliced Wasserstein algorithm on real-world data sets.** This section demonstrates the practical applicability of the sliced Wasserstein model by applying it to real-world financial data sets. This experiment is done through three steps. First, a look-ahead trading strategy is applied to assess the accuracy of the clustering results in a synthetic data type setting. Then, the focus shifts to create multiple tradeable signals from the sliced Wasserstein k-means algorithm and test it again on real-world training data sets. Finally, the section provides a discussion on the hyperparameter tuning of the trading strategies and its performance on a test data set.

- **Experiment 3: Interpretability of clustering results with macroeconomic indicators.** This section gives an insight on the relationship between the statistical clustering results of the sliced Wasserstein k-means algorithm and macroeconomic indicators. The research gives a feature importance analysis on supervised learning algorithms trained on the clustering results of the sliced Wasserstein k-means algorithm with a set of 10 macroeconomic indicators. In particular, the focus is to look at bear regime predictions of the supervised models and assess – using the SHAP analysis toolbox – which features are most important for the model.

- **Conclusion and Future Work.** In this final section, we summarize the key findings of the thesis as well as the limitations revolving around the sliced Wasserstein k-means algorithms. This will lead to a discussion of potential improvements and future research.

Chapter 2

Background and Literature Review

This chapter reviews the literature on regime detection in financial markets and positions the thesis within it, identifying the limitations that motivate the present work. Beginning with the practical importance of regime detection in the industry, it moves through parametric and structural-break methods before turning to the distributional, model-free framework on which this thesis builds. It closes by reviewing the state of Monte Carlo sampling of orthogonal projections and its role in the methodology developed here.

2.1 Importance of Regime Detection in Financial Markets

Regime detection has become over the years a heavily studied topic in literature, as it is of great interest for practitioners. In fact, the industry heavily relies on regime detection methods to adapt its trading strategies or portfolio allocation parameters to the changing dynamics of the market. Moreover, it has become crucial to scale regime identification to higher-dimensions as the market moves towards trading a basket of assets instead of just one asset.

2.2 Parametric Approaches: Hidden Markov Model

One of the first notable papers by Hamilton (1989) [7] introduced a parametric approach to regime detection using hidden Markov models (HMM). This method assumes the existence of an unobserved Markov chain that governs the parameters of the observed time-series. The HMM model is until now, one of the most widely used methods for regime detection in financial markets. Rydén et al. (1998) [17] showed that the HMMs approach reproduces most stylized facts of returns series by Granger and Ding (1995) [6] such as heavy-tails, unconditional non-normality in the returns distribution and absence of autocorrelation in raw returns. In addition, the HMM approach is a highly intuitive method for regime detection. More recently, Ang et al. (2012) [2] showed that regime in financial markets empirically switches abruptly and this changed behavior persists for many periods. This gives empirical justification to model regimes as discrete states rather than continuously varying parameters. However, HMMs struggle as the framework forces to specificy a parametric emission family of distributions that is often linked to mixture of Gaussian distributions or student-t distributions which does not necessarily align with the true empirical distribution of returns. Rydén et. al (1998) [17] further demonstrate that the HMMs models fail on the temporal dependence of absolute returns. In particular, they cannot capture the slow decay of the autocorrelation function of squared returns, which can have massive implications for risk management and portfolio allocation.

2.3 Structural Breaks Approach

Structural break approaches offer a complementary approach to HMMs for regime detection based on different assumptions. Whereas the HMMs approach assumes a fixed, finite set of regimes that recur, a structural break approach allows for a permanent non-reccuring change in the underlying dynamics of the time-series. Afterwards, parameters take new values and the previous regime is not expected to return. This is important in practice as the HMM's time invariant parameters means that a true break leaves the model with a misspecification problem rather than simply a re-estimation of parameters. Traditional regime detection approach try to identify these global beaking-points for adapting parameters such as in Bai and Perron (2003) [3] and with general-to-specific modeling approaches in Hendry et.al (2014) [9]. However, those models identify only when the breaking points occur but do not provide any insight on the underlying dynamics of the market. This is where a clustering modelling approach can be useful as it can provide through its centroid an understanding of the nature of the regime while yielding temporal segmentation of the market's behaviour.

2.4 Distributional model free Regime Detection

The Wasserstein k-means approach as proposed by Horvath et al. [10] is clustering regimes by analysing the behaviour of the *distribution* of the time-series of returns through the study period. This enables to be model-free for the returns distribution and therefore skip parameters adaptation issues. The Wasserstein k-means algorithm is the conclusion of research in optimal transport by Villani [18] and Peyré and Cuturi [15]. The Wasserstein distance on which Horvath et.al [10] relies for its algorithm has been shown to be a powerful metric for comparing distributions. Especially in the Machine Learning literature, researchers have come to understand the advantage the Wasserstein distance and Wasserstein barycenters which uses the 2-norm. In particular, their showed how those metrics satisfy convenient base properties which are crucial for comparing absolutely continuous measures (i.e., Agueh and Carlier (2011) [1] and Cuturi and Doucet (2014) [13]). In particular, the Wasserstein barycenter is unique and exists which is a crucial property for defining cluster centroids in the Wasserstein k-means algorithm [1]. In the case of empirical measures (or projected measures) in 1 dimension, the Wasserstein barycenter is a quantile average which is unique for atomic measures providing good properties for clustering distributions. Other candidates for comparing distributions include the Kullback-Leibler (KL) divergence or the Kolmogorov Smirnov Statistic. However, as expressed in Horvath et al. [10], those tools do not provide "a simple, elegant and scalable" [10] mathematical framework for clustering distributions. For instance, the KL divergence is not very well suited for working over empirical measures while the Kolmogorov Smirnov Statistic "lacks the sensitivity required to distinguish between elements of clusters" [10]. This is why the Wasserstein distance is preferred for clustering returns distributions.

The sliced Wasserstein k-means clustering algorithm as proposed by Luan et al. [11] generalizes Horvath's approach to high-dimensional time-series data and infers regimes changes using returns distribution in $\mathscr{P}_p(\mathbb{R}^d)$, the space of probability measures on $\mathbb{R}^d$ with finite p-th moment. Not only, this method adapts to high-dimensional setting, but also provides a robust and intelligible mathematical framework for regime detection. However, the sliced Wasserstein distance – on which the above algorithm is based – is an integral over the unit sphere $\mathbb{S}^{d-1}$ and therefore requires a Monte Carlo approximation to be computed in practice.

2.5 Monte Carlo Estimation and Orthogonal Sampling

While in Luan et. al [11] the authors use a fixed set of well-chosen random projection vectors to compute the sliced Wasserstein distance, authors claim that using such manually constructed projections vectors is not scalable to increasingly higher dimensional data. Rowland et al. [16] proposed a method to estimate the sliced Wasserstein distance using orthogonal projections vectors sampled from the UnifOrtho numerical method by Mezzadri (2007) [14]. This method is highly scalable to high-dimensional data and empirically provides variance reduction properties for the Monte Carlo approximation. Petrovic et al. (2025) [19] further gives the mathematical explanation behind the observed variance reduction and better sampling of the UnifOrtho method for large dimensions.

2.6 Summary

This chapter has provided key understanding of the existing literature on regime detection in financial markets. In particular, the chapter higlights how HMM models are used for regime detection but fail to capture the non-Gaussian nature of financial returns distributions. We also introduced breaking-point models as well their limitations for understanding the nature of regimes before introducing model free approaches. In particular, we focused on the literature around Wasserstein k-means clustering algorithm and how the sliced Wasserstein k-means clustering algorithm provides a robust framework for regime detection in high dimensional time-series data. Finally, we focused on the literature around the UnifOrtho method and its variance reduction properties for Monte Carlo approximation of the sliced Wasserstein distance.

Chapter 3

Methodology and Theoretical Background

This chapter develops the mathematical framework of the thesis. It begins with the Wasserstein distance and barycenter, the notions behind the Wasserstein *k*-means clustering algorithm, before presenting the sliced Wasserstein extension and its UnifOrtho refinement. We give the numerical recipe for sampling from the Haar measure on $O(d)$, yielding columns distributed as UnifOrtho($\mathbb{S}^{d-1}$), and discuss its time complexity and possible improvements. A further section examines the variance reduction UnifOrtho brings to the Monte Carlo approximation of the sliced Wasserstein distance through spherical harmonics. We conclude by deriving the implied probabilities of regime switches from the clustering results.

3.1 Introduction

This thesis uses a modified version of the k-means clustering algorithm to cluster high-dimensional empirical distributions. In particular it uses the Wasserstein distance from Horvath et al. [10] and its extension to high-dimensional distributions called the sliced Wasserstein distance (Luan and Hamp (2025) [11]) to cluster those distributions. However, understanding the key functioning of the sliced Wasserstein metric is vital to truly appreciate the regime detection approach taken in this dissertation. We therefore want to understand how the sliced Wasserstein distance mathematically detects regimes and how it can be used to predict regimes. To do so, we will first introduce the Wasserstein metric before extending its improvement using the UnifOrtho method by Rowland et al. [16]. Finally, we will introduce the mathematical framework to derive implied probabilities of regime switches from the sliced Wasserstein k-means clustering results.

3.2 Wasserstein Distance and Barycenters

To better understand the Wasserstein distance and its extension to the sliced Wasserstein distance, we will introduce a number of key concepts. In this thesis, we consider the space $\mathscr{X} = \mathbb{R}^d$ with $d \geq 1$ the dimension of the returns vector. We further define $\mathscr{P}_p(\mathbb{R}^d)$ as the set of probability measures on $\mathbb{R}^d$ with finite p-th moment. [11]

One of the key concepts introduced by Horvath et al. [10] is the notion of data streams.

Definition: Data Streams A set $\mathscr{S}$ of streams of data over $\mathscr{X}$ is defined as:

$$S(\mathscr{X}) = \{(X_1, X_2, \ldots, X_N) | X_i \in \mathscr{X}, N \in \mathbb{N}\}.$$

For instance in this research we consider the following:

$$S = (S_1, S_2, \ldots, S_N) \in \mathscr{S}(\mathscr{X}) \tag{3.1}$$

where S_i might be the the price vector of the different assets at time i. [10]

As price time-series data are known to be non-stationary, we will apply a transformation r^S to the stream S with $r^S \in \mathscr{S}(\mathscr{X}')$ defined as :

$$r_i^S = log(S_i) - log(S_{i-1}) \text{ for } i = 2, 3, \ldots, N \tag{3.2}$$

In our study $\mathscr{X} = \mathscr{X}' = \mathbb{R}^d$ and r^S is the stream of returns of the price stream S.

Definition: Lifting Transformation A lifting transformation l is a map that maps the set of streams $\mathscr{S}(\mathscr{X})$ into the set of streams of streams:

$$l : \begin{cases} \mathscr{S}(\mathscr{X}) \to \mathscr{S}(\mathscr{S}(\mathscr{X})) \\ S \mapsto l(S) = (l_1, l_2, \ldots, l_M) \end{cases} \tag{3.3}$$

where $M > 1$. [10]

The example from [4] is the sliding window transformation:

$$l^m(\mathbf{x}) = (x_{1+h_2(m-1)}, x_{2+h_2(m-1)}, \ldots, x_{h_1+h_2(m-1)}) \text{ for } m = 1, 2, \ldots, M \tag{3.4}$$

with $M := \left\lfloor \frac{N-(h_1-h_2)}{h_2} \right\rfloor$ and $h_1, h_2 \in \mathbb{N}$ the window size and the step size respectively. This transformation is particularly useful as it defines the samples on which on defines the empirical discrete distribution μ^S which we define below.

Definition. Discrete uniform distribution. Given a stream generated by the lifting process $l(S) = (l^m(\mathbf{x})_{i,} : i = 1, 2, \dots, h_1)$, we can define the discrete uniform distribution μ^S as follows:

$$\mu_m := \frac{1}{h_1} \sum_{i=1}^{h_1} \delta_{l^m(\mathbf{x})_i} \tag{3.5}$$

where we have that δ_x is the Dirac delta function at point x. [11] This definition allows us to introduce the set of empirical distributions $\mathscr{K} = \{\mu_j\}_{1 \le j \le M}$ where $\mu_j \in \mathscr{P}_p(\mathbb{R}^d)$ with $\mathscr{P}_p(\mathbb{R}^d)$ the set of probability measures on $\mathbb{R}^d$ with finite p-th moment. [11].

Definition. Wasserstein distance $\mathscr{W}_p$. Let μ and ν be two probability measures on $\mathbb{R}^d$ with finite p-th moment. The Wasserstein distance of order p between μ and ν is defined as follows:

$$\mathscr{W}_p(\mu, \nu) := \inf_{(X,Y) \sim \Gamma(\mu,\nu)} (\mathbb{E}[\|X - Y\|^p])^{1/p} = \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\pi(x, y) \right)^{1/p} \tag{3.6}$$

where $\Gamma(\mu, \nu) \subseteq \mathscr{P}_p(\mathbb{R}^d \times \mathbb{R}^d)$ is the set of joint distributions over $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν. An element $\pi^* \in \Gamma(\mu, \nu)$ that achieves the infimum is called an optimal transport plan between μ and ν. [16]

Definition. Wasserstein barycentre $\bar{\mu}$. Given a family probability measures $\mathscr{K}$ in the space $\mathbb{R}^d$ with finite p-th moment, the Wasserstein barycentre $\bar{\mu}$ is defined as the probability measure that minimizes the total Wasserstein distances to the members of the family $\mathscr{K}$:

$$\bar{\mu} := \arg \min_{\nu \in \mathscr{P}_p(\mathbb{R}^d)} \sum_{\mu_m \in \mathscr{K}} \mathscr{W}_p(\nu, \mu_m) \tag{3.7}$$

Projected Empirical Measure. Let $\theta \in \mathbb{S}^{d-1}$ with $\mathbb{S}^{d-1}$ the unit sphere in d dimensions. The projected empirical measure $\mu'(\theta) \in \mathscr{P}_p(\mathbb{R})$ is defined as:

$$\mu'(\theta) := \frac{1}{N} \sum_i \delta_{\langle x_i, \theta \rangle} \tag{3.8}$$

with x_i the atoms of the original empirical measure $\mu \in \mathscr{P}_p(\mathbb{R}^d)$ and $\langle \cdot, \cdot \rangle$ the standard inner product in $\mathbb{R}^d$. [11]

Definition. The sliced Wasserstein distance $\tilde{\mathscr{W}_p}$. Given two distributions $\mu, \nu \in \mathscr{P}_p(\mathbb{R}^d)$ with $d > 1$, the sliced Wasserstein distance is defined as follows:

$$\widetilde{\mathscr{W}_p}(\mu, \nu) := \int_{\mathbb{S}^{d-1}} \mathscr{W}_p(\mu'(\theta), \nu'(\theta)) d\theta \tag{3.9}$$

with $\mathscr{W}_p(\mu'(\theta), \nu'(\theta))$ the Wasserstein distance in d=1 between the projected empirical measures $\mu'(\theta)$ and $\nu'(\theta)$ on the direction θ. [11]

In practice one can approximate the sliced Wasserstein distance using Monte Carlo methods by sampling L random directions $\{\theta^1, \ldots, \theta^L\}$ uniformly on the unit sphere $\mathbb{S}^{d-1}$ and computing the average Wasserstein distance over these projections:

$$\widetilde{\mathscr{W}_p}(\mu, \nu) \approx \frac{1}{L} \sum_{l=1}^{L} \mathscr{W}_p(\mu'(\theta^l), \nu'(\theta^l)) \tag{3.10}$$

More precisely we have, $\theta^1, \ldots, \theta^L \sim^{i.i.d.}$ Unif($\mathbb{S}^{d-1}$). [16] One of the thesis primary focus is to improve the sampling from Unif($\mathbb{S}^{d-1}$) with respect to the work by Luan and Hamp. [11].

Definition. The sliced Wasserstein Barycenter. The sliced Wasserstein barycenter $\bar{\mu}^{\widetilde{\mathscr{W}_p}}$ of a family of probability measures $\mathscr{K} = \{\mu_j\}_{1 \leq j \leq M}$ is defined as:

$$\bar{\mu}^{\widetilde{\mathscr{W}_p}} := \arg \min_{\nu \in \mathscr{P}_p(\mathbb{R}^d)} \sum_{\mu_m \in \mathscr{K}} \widetilde{\mathscr{W}_p^p}(\nu, \mu_m) \tag{3.11}$$

[11]

This barycenter will be used in the sliced Wasserstein k-means clustering algorithm for representing the centroids of the clusters.

3.3 Sliced Wasserstein distance and UnifOrtho Projections

This section will provide insights on the mathematical background needed to understand the UnifOrtho approach for sampling projections vectors for the sliced Wasserstein distance. Notably, we will see that this method solves the low generalizable nature of using pre-defined projection vectors θ^l for the sliced Wasserstein distance as proposed by Luan et al [11]. Instead the Unifortho approach provides a more robust and generalizable clustering algorithm in higher-dimensions.

As we have seen in the previous section, the sliced Wasserstein distance is approximated using L sampling of random direction in the unit sphere $\mathbb{S}^{d-1}$ for a Monte Carlo estimation of the distance. This UnifOrtho method relies on the properties of the orthogonal group $O(d)$ and its associated Haar measure to sample those directions so that they are orthogonal two by two.

Before introducing the UnifOrtho method let us first consider the following definitions.

Definition. Orthogonal group $O(d)$. The orthogonal group $O(d)$ is defined as :

$$O(d) = \{U \in \mathbb{R}^{d\times d} \mid U^T U = I_d\} \tag{3.12}$$

Definition. Haar measure on $O(d)$. The Haar measure on $O(d)$ is the unique probability measure that satisfies the following properties:

1. $\mu(QA) = \mu(A)$ for all $Q \in O(d)$ and Borel sets $A \subseteq O(d)$ (left-invariance)
2. $\mu(AQ) = \mu(A)$ for all $Q \in O(d)$ and Borel sets $A \subseteq O(d)$ (right-invariance)
3. $\mu(O(d)) = 1$ (normalization)

This allows us to introduce the UnifOrtho distribution which is the key concept behind the UnifOrtho method for sampling orthogonal projections vectors for the sliced Wasserstein distance.

Definition. UnifOrtho$(\mathbb{S}^{d-1}, N)$ Let $N \leq d$ be a positive integer. The probability distribution UnifOrtho$(\mathbb{S}^{d-1}, N) \in \mathscr{P}((\mathbb{S}^{d-1})^N)$ is defined as the joint distribution of N rows of a random matrix drawn from the Haar distribution on the orthogonal group $O(d)$. In other words, if U is a random matrix drawn from the Haar distribution on $O(d)$, then the distribution of any N rows of U is UnifOrtho$(\mathbb{S}^{d-1}, N)$. [16]

Numerical Recipe for constructing matrices with Haar distribution on $O(d)$ from Mezzadri (2007) [14]

1. Take a $d \times d$ complex matrix Z whose entries are complex standard normal random variables.
2. Feed Z into *any* QR decomposition routine. Let (Q, R), where $Z = QR$, be the output.
3. Create the following diagonal matrix

$$\Lambda = \begin{pmatrix} \frac{r_{11}}{|r_{11}|} & & \\ & \ddots & \\ & & \frac{r_{dd}}{|r_{dd}|} \end{pmatrix}, \tag{3.13}$$

   where the $\{r_{jj}\}_{1\leq j\leq d}$ are the diagonal elements of R.

4. The diagonal elements of $R' = \Lambda^{-1}R$ are *always* real and strictly positive, therefore the matrix $Q' = Q\Lambda$ is distributed with Haar measure.

Hence we have that $Q' \sim \text{Haar}(O(d))$ and its columns $q'_i \in \text{Unif}(\mathbb{S}^{d-1})$ for $i = 1, \ldots, d$ with any $N \leq d$ columns of Q' being jointly distributed according to $\text{UnifOrtho}(\mathbb{S}^{d-1}; N)$.

Further explanation on understanding why the above recipe yields uniformly distributed orthogonal matrices in $O(d)$ can be found in [14]. In our work we will not be sampling Z from a complex normal distribution but rather from a real normal distribution.

Combining the above numerical schema [16, 14] as well as the general sliced Wasserstein k-means approach [11] we devise therefore the following pseudo code (Algorithm 1). Below *sgn* refers to the sign function which is the equivalent normalization of the diagonal elements of the matrix R for real numbers.

Algorithm 1: sWk-means clustering with UnifOrtho Projections

Input: Data stream $S \in \mathscr{S}(\mathbb{R}^d)$,;
number of clusters K,;
number of projections L,;
tolerance ε for convergence
Output: K cluster assignments and corresponding 1d barycenters

▷ Preprocessing
Calculate $l(r^S)$ given $S \in \mathscr{S}(\mathbb{R}^d)$ and define family of empirical distributions $\mathscr{K} = \{\mu_j\}_{1 \leq j \leq M}$;

▷ Step 1: UnifOrtho Projection Generation
$k \leftarrow \lceil L/d \rceil$; ▷ number of orthogonal bases needed
$\Theta \leftarrow \emptyset$;
for $i = 1$ **to** k **do**
  Sample a random matrix $Z_i \in \mathbb{R}^{d \times d}$ with i.i.d. $\mathscr{N}(0,1)$ entries;
  Compute the QR decomposition: $Z_i = Q_i R_i$;
  Let Λ_i be a diagonal matrix where $\Lambda_{i,jj} = \mathrm{sgn}(R_{i,jj})$;
  $U_i \leftarrow Q_i \Lambda_i$; ▷ U_i is Haar-distributed on $O(d)$
  Add the d column vectors of U_i to Θ;
$\{\theta^1, \dots, \theta^L\} \leftarrow \Theta[1:L]$; ▷ select exactly L directions

▷ Step 2: Projection & K-Means Iteration
Calculate projected distributions $\{\mu'_j(\theta^l)\}_{1 \leq j \leq M}$ for all $l = 1 \dots L$;
Initialize cluster assignments $C_1, \dots, C_K$;
while *convergence criterion* $> \varepsilon$ **do**
  for $k = 1$ **to** K **do**
    Compute the 1d Wasserstein barycenter $\bar{\mu}_k(\theta^l)$ for each projection l using empirical distributions assigned to C_k;
  for $j = 1$ **to** M **do**
    Assign μ_j to the cluster C_{k^*} where $k^* = \arg\min_k \sum_{l=1}^{L} W_p^p\left(\mu'_j(\theta^l), \bar{\mu}_k(\theta^l)\right)$;
return $C_1, \dots, C_K$ *and barycenters* $\{\bar{\mu}_k(\theta^l)\}$

This sliced Wasserstein estimation method has a polynomial computational time complexity. Indeed, the UnifOrtho projection generation step alone is $\mathscr{O}(d^3)$ due to the QR decomposition of k matrices of size $d \times d$. In addition, the projection and k-means iteration step has a complexity of $\mathscr{O}(MKL)$ for the assignment step and $\mathscr{O}(KLM_k \log M_k)$ for computing the 1d Wasserstein barycenters. Here M_k refers to the number of distributions assigned to cluster C_k, K to the numbers of clusters and L to the number of projections. One improvement that could be done is to use the Hadamard product, for reducing the time complexity to

$\mathscr{O}(d^2 \log d)$ and for approximatively sampling from UnifOrtho($\mathbb{S}^d; N$) generating the UnifOrtho projections by leveraging structured random matrices but this is beyond the scope of this work [5].

In parallel, in the above algorithm we introduce a notion of *convergence criterion* without explicitly defining it. This convergence criterion will be defined as in Luan and Hamp (2025) [11] as:

$$\sum_k \tilde{\mathscr{W}}_p\left(\bar{\mu}_k^n, \bar{\mu}_k^{n-1}\right) < \varepsilon \tag{3.14}$$

where $\bar{\mu}_k^n$ is the centroid of cluster C_k at the n-th iteration. In our research we will be setting $\varepsilon = 10^{-6}$.

3.4 Variance of UnifOrtho Estimator

As shown in Rowland et al. (2019) [16], the variance of the Monte Carlo estimator of the sliced Wasserstein distance using UnifOrtho projections is not always a guarantee of variance reduction. In fact, Rowland et al. (2019) [16] shows that there are cases in which the variance of the UnifOrtho are worse than an i.i.d Monte Carlo estimator. However, as shown in Petrovic et al. (2025) [19], the nature of the Wasserstein distance, and the fundamental properties of spherical harmonics, explain why the UnifOrtho estimator showcases lower variance for higher dimensions $d \geq 3$.

Proposition Let f be a continuous function on $\mathbb{S}^{d-1}$, and $(X_1 | \dots | X_d)$ be a matrix drawn from the Haar measure on the orthogonal group $O(d)$. Let Y_k^ℓ, $\ell \geq 0$, $1 \leq k \leq h_\ell$ be a basis of spherical harmonics, and $\hat{f}(\ell, k) = \int_{\mathbb{S}^{d-1}} f(x) \mathrm{Y}_k^\ell(x)\, \mathrm{d}x$ denote the spherical coefficients of f. Then

$$\begin{aligned} \mathrm{Var}\left(\frac{1}{d}\sum_{i=1}^{d} f(X_i)\right) &= \frac{1}{d}\mathrm{Var}(f(X_1)) - \frac{d-1}{d}\sum_{\ell=1}^{+\infty}(-1)^{\ell-1}\lambda_{2\ell}\mu_{2\ell}(f) &(3.15)\\ &= \frac{1}{d}\mathrm{Var}(f(X_1)) - \frac{d-1}{d}\sum_{\ell=1}^{+\infty}\lambda_{4\ell-2}\left(\mu_{4\ell-2}(f) - \alpha_{2\ell-1}\mu_{4\ell}(f)\right), &(3.16)\end{aligned}$$

where

$$\mu_\ell(f) = \sum_{k=1}^{h_\ell} \hat{f}(\ell,k)^2, \quad \alpha_\ell = \frac{2\ell+1}{2\ell+d-1}, \quad \textit{and } \lambda_{2\ell} = \frac{\Gamma\left(\frac{d-1}{2}\right)\Gamma\left(\frac{2\ell+1}{2}\right)}{\sqrt{\pi}\,\Gamma\left(\frac{2\ell+d-1}{2}\right)}.$$

[19]

As explained in [19] we see that the variance of the UnifOrtho estimator in 3.15 and 3.16 is reduced or increased depending on the "energy profile" of $(\mu_{2\ell}(f))_{\ell\in\mathbb{N}}$ of the integrand f.

One of the key insights shown by [19] is that for large values of d, we can use the generalized Stirling formula to approximate $\lambda_{2\ell} = \mathcal{O}(\ell^{-(d-1)/2})$. This yields that only the first terms in the alternating series in 3.15 and 3.16 is significant. This pulls the variance of the UnifOrtho estimator downwards and explains why the UnifOrtho estimator is more efficient in higher dimensions in general. This should not be taken as a proof but rather as an intuitive explanation of the experimental variance reduction results of UnifOrtho observed in [16] and [19].

3.5 Implied Probability of Regime Switches

This thesis in addition to providing improvements on Luan and Hamp (2025) [11] regime detection method also investigates the use of the sliced Wasserstein distance to derive tradeable signals and predictions of future regime switches in the market.

3.5.1 Bayesian Inference for Regime Switches

One major improvement this thesis proposes is the construction of likelihood function for determining the probability of being in a certain regime given the current empirical distribution of returns. In particular, we use the sliced Wasserstein distances of a fixed empirical distribution x_m to every centroid. Then, one converts this to a likelihood function via a softmax over the negative distances as shown below. We introduce a temperature parameter τ for controlling the sharpness of the likelihood function.

$$\mathcal{L}(\text{regime } k \mid x_m) := \frac{e^{-d(x_m,c_k)/\tau}}{\sum_j e^{-d(x_m,c_j)/\tau}} \tag{3.17}$$

with

$$x_m := \frac{1}{N}\sum_{i=1}^{N} \delta_{l^m(\mathbf{x})_i}$$

the empirical distribution of the current window in $\mathscr{P}_p(\mathbb{R}^d)$ with a lifting transformation. After having run the sliced Wasserstein k-means clustering algorithm, on the target window of the data, we have for $k = 1,\dots,K$:

$$c_k^* = \arg\min_{\nu\in\mathscr{P}_p(\mathbb{R}^d)} \sum_{\mu_m\in C_k} \widetilde{W}_p^p(\nu,\mu_m), \quad k = 1,\dots,K \tag{3.18}$$

and we have for ν and μ in $\mathscr{P}_p(\mathbb{R}^d)$:

$$d(\nu,\mu) := \widetilde{\mathscr{W}_p}(\nu,\mu)$$

To construct a predictive signal we compute empirical transition matrix π using a first-order Markov Chain with the historical labels of the regimes. Mathematically, one can define the probability of switching from regime i to regime j as follows:

$$\pi_{i,j} := P(Z_m = j \mid Z_{m-1} = i) = \frac{\text{Count}(i \to j)}{\sum_{x=1}^{K} \text{Count}(i \to x)} \tag{3.19}$$

If the market is currently at state z_{current}, the prior probability of transitioning to regime k is given by the corresponding row of the transition matrix

$$\mathbb{P}_{prior}(k) := \pi_{z_{current},k} \tag{3.20}$$

We then apply Bayes rule to update this prior with the likelihood of the current observation X_m yielding the posterior probability:

$$\mathbb{P}_{Bayes}(k) = \frac{\mathbb{P}_{prior}(k) \cdot \mathscr{L}(k \mid X_m)}{\sum_{j=1}^{K} \mathbb{P}_{prior}(j) \cdot \mathscr{L}(j \mid X_m)} \tag{3.21}$$

We then apply simple probability principle to derive the probability of switching from the current regime to any other regime as follows:

$$\mathbb{P}_{switch} = 1 - \mathbb{P}_{Bayes}(z_{current}) \tag{3.22}$$

3.5.2 Momentum for Regime Switches

Another consideration when computing switching probability is to take into account the momentum of regimes. The underlying modelling hypothesis is whether the market is drifting away from the current regime or is it confirming the current regime. Mathematically, we translate this by considering a lookback window of size ω. One then computes the linear trend slope of the Wasserstein distance to each centroid k as follows. Let $t \in \{1, 2, \ldots, \omega\}$ and using OLS, β_k we get:

$$\beta_k := \frac{\text{Covariance}(t, \mathscr{W}_2(X_{M-w+t}, C_k))}{\text{Variance}(t)} \tag{3.23}$$

- if $\beta_k < 0$ the distance to regime k is shrinking (we are getting close to regime k)
- if $\beta_k > 0$ the distance to regime k is growing (we are drifting away from regime k)

Again, we apply a softmax function to get a "stay" probability based on the magnitude and sign of β_k. We also introduce a new temperature parameter τ_{grad}.

$$P_{grad}(k) := \frac{\exp\left(-\frac{\beta_k}{\tau_{grad}}\right)}{\sum_{j=1}^{K} \exp\left(-\frac{\beta_j}{\tau_{grad}}\right)} \tag{3.24}$$

As we can see in 3.24, the more negative the β_k, the higher the $exp(-\beta_k)$ expression and the more likely we are to stay in regime k. Inversely, the more positive the β_k and the more likely we are to switch away from regime k.

The variant introducing momentum of regime is therefore a weighted average with the Bayes probability and expresses itself as follows:

$$\mathbb{P}_{switch} = 1 - [(1-\alpha)\mathbb{P}_{Bayes}(z_{current}) + \alpha\mathbb{P}_{Grad}(z_{current})] \tag{3.25}$$

with $\alpha \in [0,1]$ a weight that can be tuned on a validation set. The tuning of α is interesting because it implicitly indicates whether the predictive signal for regime switches is more driven by momentum or by the distance to the centroids.

3.6 Summary

In this chapter, we layed the ground foudations for understanding the sliced Wasserstein k-means algorithm from the Wasserstein distance to its barycenter. We also introduced the UnifOrtho projection method and numerical recipe for generating orthogonal projection vectors from the Haar measure on the orthogonal group $O(d)$. This allowed us to show how the UnifOrtho sampling method intuitively reduces variance by studying the spherical harmonics of the variance of the Monte Carlo estimator. Finally, we introduced a method for deriving predictive signals for regime switches based on the sliced Wasserstein distance to the centroids. Additionally, we introduce a first-order Markov Chain model for the transition probabilities between regimes.

Chapter 4

Experiment 1: Synthetic Data Testing

This chapter tests the sliced Wasserstein method on synthetic data to characterise the algorithm's behaviour across a range of controlled market scenarios. We evaluate the sliced Wasserstein *k*-means algorithm under a variety of tuning parameters, finding consistently high performance across most settings. Benchmarking against Luan and Hamp (2025) [11], we show that the UnifOrtho projections improve overall accuracy. We then examine individual runs under fixed tunings and, looking beyond clustering accuracy to balanced accuracy, confirm that the model performs well across the majority of scenarios considered.

4.1 Introduction

This dissertation focuses on a novel approach to detect regimes and therefore has to be tested on different market scenarios. As a first benchmark, it is highly relevant to generate a wide variety of synthetic data to simulate those different market conditions. The chapter therefore focuses on comparing the sliced Wasserstein k-means algorithm using UnifOrtho with the results in the literature on synthetic data (Luan and Hamp (2025) [11]) to see if it provides better accuracy. We will first test the novel algorithm on different synthetic data settings with a range of tunings as in Luan and Hamp (2025) [11]. We then plot single runs of the sliced Wasserstein algorithm with fixed tuning on the different synthetic data sets. The goal is to get a good understanding of the behavior of our modelling approach in different artificial market scenarios.

4.2 Background

In Luan and Hamp (2025) [11], the authors introduce the sliced Wasserstein k-means clustering algorithm to detect regimes in financial time series data and test the algorithm on synthetic data for 2-dimensional data and 3-dimensional data. The paper concludes that the sliced Wasserstein k-means clustering algorithm has an increasing accuracy as one elevates the number of projections L and the window size h_1. Before we present the results, let us first detail the accuracy metrics used in the reference experiment in Luan and Hamp (2025) [11]. In this experiment, the authors use mainly two metrics to evaluate the performance of the clustering algorithm in the synthetic data setting:

- Total Accuracy (TA)
- Mean centroid-centroid distance (MCCD)

.

Definition. Total Accuracy. The Total Accuracy (TA) as defined in [11] is the proportion of correctly classified data points over the total number of data points. Mathematically, with each data returns points $r_{t_i}^S \in \mathscr{X}$ is assigned a label $\hat{y}_{t_i} \in \{1,\ldots,K\}$. The Total Accuracy achieved by the clustering $\mathscr{C} = \{\hat{y}_{t_i}\}_{i=1}^N$ is defined as follows:

$$\mathrm{TA}(\mathscr{C}) := \frac{\sum_{i=1}^N \mathbb{I}_{\hat{y}_{t_i}=y_{t_i}}}{\sum_{i=1}^N (\mathbb{I}_{\hat{y}_{t_i}=y_{t_i}} + \mathbb{I}_{\hat{y}_{t_i}\neq y_{t_i}})} \tag{4.1}$$

where y_{t_i} is the true label of the data point $r_{t_i}^S$ and $\mathbb{I}$ is the indicator function.

Definition. Mean Centroid-Centroid Distance (MCCD). Let the centroids of the clusters be denoted as $\{\bar{\mu}_k\}_{k=1}^K$, the Mean Centroid-Centroid Distance (MCCD) is defined as follows:

$$\langle \mathscr{W}_p(\bar{\mu}_k, \bar{\mu}_{k'}) \rangle_{k,k'} := (C_2^K)^{-1} \sum_{k' \neq k} \mathscr{W}_p(\bar{\mu}_k, \bar{\mu}_{k'}), \tag{26}$$

where $C_2^K = \binom{K}{2} = \frac{K(K-1)}{2}$ is the number of unique pairs of clusters. This metric represents how well-separated the different clusters are from each other in the $\mathscr{P}_p(\mathbb{R}^d)$ space. A higher MCCD indicates that the clusters are more distinct and less likely to overlap, which is desirable for effective clustering. Later for real datasets (i.e in an unsupervised setting) this metric will be used to evaluate the quality of the clustering as there will be no ground truth labels available for computing the Total Accuracy (TA). It is therefore important to consider how well the MCCD metric correlates with the Total Accuracy (TA) in the synthetic setting to understand whether it can be used as a proxy for evaluating clustering performance in real-world scenarios.

4.3 Data Set

The sliced Wasserstein k-means algorithm will be tested on 2, 3, 4, 5 and 100 dimensional synthetic data. Performance of the UniOrtho method will be evaluated with data including multiple correlation factors and with different target number of clusters. The synthetic data used in this experiment is the equivalent of 20 years of data with daily frequency. Therefore, it comprises $20 * 252 = 5040$ data points. We will generate this synthetic data similarly to previous papers [10] and [11] using Geometric Brownian Motion (GBM) paths $S_t/S_{t-1} = \exp(r_t^S)$ with $S_t = (S_t^1, \ldots, S_t^d)$ and $r_t^S = (r_t^{S^{(1)}}, \ldots, r_t^{S^{(d)}})$ where :

$$r_t^{S^{(i)}} \sim \mathcal{N}\left((\mu - \sigma^2/2)\mathbf{1}dt, \Sigma dt\right) \tag{4.2}$$

. with

$$\Theta = (\mu, \sigma^2) \tag{4.3}$$

The covariance matrix Σ is defined as follows :

$$\Sigma_{ij} = \sigma^2 \left(\delta_{ij} + (1 - \delta_{ij})\rho\right) \tag{4.4}$$

The δ_{ij} is the Kronecker delta, σ^2 is the variance of the returns and ρ is the correlation between the different assets.

Assuming, we want to cluster two different regimes (K=2), we define, as in previous papers [11], *bull* and *bear* regimes with the following parametrization :

- $\Theta_{\text{bull}} = (0.02, 0.2)$
- $\Theta_{\text{bear}} = (-0.02, 0.3)$

For K = 3, similarly to Luan and Hamp (2025) [11] we will consider the *moon-shape structure* generated using `datasets.make_moons` function from the `sklearn` library.

Table 4.1 presents the different market settings on which we are going to test the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections :

Table 4.1: Summary of synthetic dataset parameters with $d \in \{2, 3, 4, 5, 100\}$ and with 5040 data points.

Type	Regime Bull	Regime Bear	Regime III (Bear/Moon)
Type A (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	N/A
Type B (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bull}}; \rho = -1/2$	N/A
Type C (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = -1/2$
Type D (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{moon}}; \rho = +1/2$

4.4 Results and Discussion

As in Luan and Hamp (2025)[11] we have tested our sliced Wasserstein k-means clustering algorithm with UnifOrtho projections on the above mentioned synthetic data as well as with different tunings of the parameters of h_1, h_2 and L (resp. the window size, the step size and the number of projections). For purposes of brevity, we present in this chapter, only results for the case of $d = 5$. However, interested readers can find results for $d = 2, 3, 4, 100$ in the appendix. Each results in table is either the maximum total accuracy (TA), the median total accuracy and the total accuracy of the run with the maximum mean centroid-centroid distance (MCCD) over $N_S = 50$ runs of the sliced Wasserstein algorithm.

4.4.1 Type A and B data

Table 4.2 presents the results of the sliced Wasserstein algorithm with the UnifOrtho projections across parameters h_1, h_2, L for type A and B data as presented in table 4.1

h_1	h_2	L	Max		Median		Max(MCCD)	
			A	B	A	B	A	B
20	4	9	0.9890	0.9985	0.9888	0.9985	0.9889	0.9985
		20	0.9896	0.9983	0.9895	0.9982	0.9895	0.9982
		40	0.9908	0.9986	0.9908	0.9986	0.9907	0.9986
		100	0.9889	0.9986	0.9888	0.9985	0.9888	0.9986
30	6	9	0.9950	0.9980	0.9950	0.9980	0.9949	0.9980
		20	0.9949	0.9984	0.9948	0.9983	0.9949	0.9983
		40	0.9951	0.9984	0.9948	0.9983	0.9948	0.9984
		100	0.9951	0.9982	0.9949	0.9982	0.9951	0.9982
40	8	9	0.9966	0.9981	0.9964	0.9980	0.9964	0.9979
		20	0.9965	0.9982	0.9964	0.9980	0.9963	0.9980
		40	0.9964	0.9982	0.9962	0.9980	0.9962	0.9980
		100	0.9968	0.9982	0.9967	0.9980	0.9966	0.9980
50	10	9	0.9965	0.9969	0.9962	0.9966	0.9965	0.9966
		20	0.9973	0.9974	0.9971	0.9973	0.9971	0.9972
		40	0.9973	0.9973	0.9971	0.9971	0.9971	0.9973
		100	0.9971	0.9973	0.9971	0.9973	0.9968	0.9973
60	12	9	0.9965	0.9963	0.9962	0.9962	0.9962	0.9960
		20	0.9960	0.9962	0.9957	0.9959	0.9960	0.9959
		40	0.9963	0.9965	0.9962	0.9964	0.9963	0.9965
		100	0.9960	0.9962	0.9957	0.9962	0.9957	0.9962

Table 4.2: Synthetic Type A/B (K=2), dimension d=5: Maximum total accuracy (MAX) and Maximum Mean-Centroid-Centroid Distance (MCCD) of the SW k-means clustering with UnifOrtho projections over 50 runs across window size h_1, step h_2 and number of projections $L \in \{9, 20, 40, 100\}$. Dark red $\approx$ 1, pale $\approx$ 0.5.

In the table above we observe that for data type A and B, the UnifOrtho projections yield very high accuracy and high total accuracy associated to MCCD for all tested hyperparameter settings. The results show that the UnifOrtho projections yield even better results than predefined projections in [11] with lower dimensions ($d = 2, 3$) for identical (h1, h2 and L). For instance, in Luan and Hamp (2025) [11] for data type A and B with $d = 2$ and parametrization $(h_1, h_2) \in \{(20, 4), (30, 6)\}$ the algorithm only yields median total accuracy TA($\mathscr{C}$) ranging from 0.52 to 0.725 with $N_S = 100$ clustering runs. This contrasts with the results in Table 4.2 with its stable numbers across parametrization above 0.99 for a lower number of runs $N_S = 50$. Curious readers can also find that we observe similar results for $d = 2, 3, 4, 100$ in the appendix in which for $h1 \geq 20$, total accuracy metrics and MCCD remain consistently high for type A and B synthetic data. As found in Luan and Hamp (2025)[11], we observe an improvement in total accuracy as one increases the window size h_1, however such improvement is not as significant here. In fact, we observe that the total accuracy remains consistently high across all tested window sizes h_1. This is a strong indication that the UnifOrtho projections not only is more efficient in higher dimensions but also generates more stable and accurate results across tested tunings.

4.4.2 Type C and D data

Table 4.3 presents the results of the sliced Wasserstein algorithm with the UnifOrtho projections across parameters h_1, h_2, L for type C and data as presented in table 4.1.

h_1	h_2	L	Max		Median		Max(MCCD)	
			C	D	C	D	C	D
20	4	9	0.9904	0.8633	0.5264	0.5264	0.9904	0.8633
		20	0.9915	0.8630	0.5208	0.5235	0.9915	0.8629
		40	0.9912	0.8670	0.5207	0.5220	0.9912	0.8670
		100	0.9936	0.8640	0.5244	0.5196	0.9934	0.8631
30	6	9	0.9964	0.8768	0.5073	0.5254	0.9964	0.8768
		20	0.9965	0.8767	0.5128	0.5307	0.9954	0.8767
		40	0.9968	0.8788	0.9967	0.5259	0.9968	0.8788
		100	0.9970	0.8789	0.5137	0.5374	0.9970	0.8788
40	8	9	0.9959	0.8793	0.5206	0.5269	0.9949	0.8793
		20	0.9967	0.8797	0.9951	0.5365	0.9959	0.8797
		40	0.9973	0.8794	0.7599	0.5230	0.9973	0.8792
		100	0.9973	0.8817	0.9953	0.5319	0.9956	0.8813
50	10	9	0.9943	0.8815	0.5387	0.5301	0.9930	0.8805
		20	0.9959	0.5311	0.7698	0.5241	0.9954	0.5302
		40	0.9965	0.8825	0.5503	0.5266	0.9942	0.8825
		100	0.9965	0.8799	0.5446	0.5299	0.9959	0.8799
60	12	9	0.9961	0.8814	0.7672	0.5289	0.9961	0.8814
		20	0.9959	0.8795	0.9938	0.5318	0.9956	0.8795
		40	0.9959	0.8807	0.7587	0.5357	0.9959	0.8805
		100	0.9963	0.5401	0.7630	0.5362	0.9963	0.5357

Table 4.3: Synthetic Type C/D (K=3), dimension d=5: Maximum total accuracy (MAX) and Maximum Mean-Centroid-Centroid Distance (MCCD) of the SW k-means clustering with UnifOrtho projections over 50 runs across window size h_1, step h_2 and number of projections $L \in \{9, 20, 40, 100\}$. Dark red $\approx$ 1, pale $\approx$ 0.5.

In table 4.3, we observe that for data type C and D, the UnifOrtho struggles significantly more than for data type A and B. While maximum total accuracy remains very high for data type C, the median total accuracy is mostly around 50 % which indicates that the algorithm is not able to consistently classify the data points correctly. We observe in this context that the hyperparameter tuning of h_1, h_2 and L has also a low impact on the performance of the algorithm, with higher values of h_1 giving in general only slightly better results. While the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections performs relatively well on GBM data even for $K = 3$, it struggles when faced with more complex patterns such as the moon-shape structure in type D. We observe that the median total accuracy for type D data is consistently around 50% across all tested hyperparameter settings, while providing relatively low Total Accuracy (TA) (86-88%), indicating that the algorithm is highly unstable for this type of data. This unstable behavior is also shown as the maximum total accuracy is around 54% with $h_1 = 60$ and $L = 100$ which is very low compared to the maximum total accuracy for type C data (99.63%).

A final interesting observation is that the MCCD metric is steadily very close to the maximum total accuracy metric for all types of synthetic data (i.e even for type D data) on which the sliced Wasserstein k-means algorithm is tested. Those results improve the findings by Luan and Hamp (2025) [11], who finds that there are some discrepancies between the MCCD metric and the maximum total accuracy metric for small values settings in type D data.

4.4.3 Analysis with fixed hyperparameter settings

In figure 4.1, 4.2, 4.3 and 4.4 we show the different regime detection plots for the different types of synthetic data (A, B, C and D) with fixed parameters $d = 5$, $h_1 = 30$, $h_2 = 6$ and $L = 20$. The plots reveal the regime detection results generated by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections by taking the maximum of total accuracy across $N_S = 50$ runs. In the caption of the plots, we provide the maximum total accuracy achieved by the algorithm as well as a balanced accuracy metric from `scikit-learn` to account for the imbalance in the number of data points corresponding to each regime. Red periods represent bear regimes and green periods the bullish ones. The yellow periods indicate either the bear regime with negative correlation for type C or the moon-shape structure for type D as indicated in table 4.1.

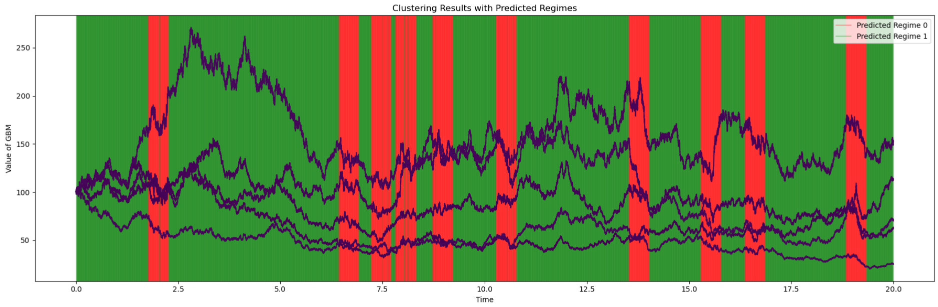


Figure 4.1: Regime detection plot for synthetic data type A with $d = 5$, $h_1 = 30$, $h_2 = 6$, and $L = 20$. The max total accuracy detected regimes by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections with $N_S = 50$. Total Accuracy(TA) = 0.9954, Balanced Accuracy(BA) = 0.9920, Mean Centroid-Centroid Distance(MCCD) = 0.0021.

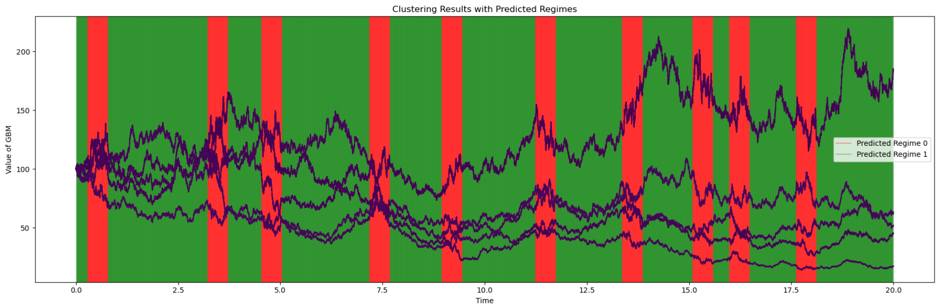


Figure 4.2: Regime detection plot for synthetic data type B with $d = 5$, $h_1 = 30$, $h_2 = 6$, and $L = 20$. The max total accuracy detected regimes by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections with $N_S = 50$. Total Accuracy(TA) = 0.9984, Balanced Accuracy(BA) = 0.9986, Mean Centroid-Centroid Distance(MCCD) = 0.0038.

We observe that the proposed regime detection method is able to detect the regimes in synthetic data type A and B with very high accuracy as well as high balanced accuracy. This is consistent with the results in Table 4.2 and shows that the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections is able to detect regimes in synthetic data with high accuracy and stability across different hyperparameter settings.

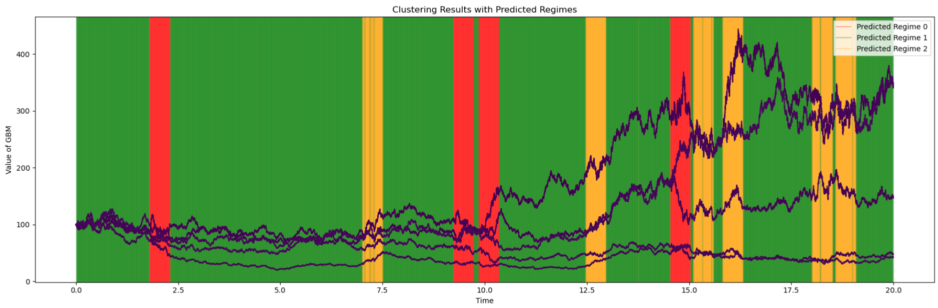


Figure 4.3: Regime detection plot for synthetic data type C with $d = 5$, $h_1 = 30$, $h_2 = 6$, and $L = 20$. The max total accuracy detected regimes by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections with $N_S = 50$. Total Accuracy(TA) = 0.9957, Balanced Accuracy(BA) = 0.9926, Mean Centroid-Centroid Distance(MCCD) = 0.0030.

Figure 4.4: Regime detection plot for synthetic data type D with $d = 5$, $h_1 = 30$, $h_2 = 6$, and $L = 20$. The max total accuracy detected regimes by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections with $N_S = 50$. Total Accuracy(TA) = 0.8745, Balanced Accuracy(BA) = 0.6732, Mean Centroid-Centroid Distance(MCCD) = 0.0020.

While one observe particularly high accuracy and balanced accuracy for type C, we observe that type D struggles with only 87% total accuracy and 67% balanced accuracy. This is particularly noticeable when plotting the regime predictions for type D in Figure 4.4 where we observe that the regimes are visually not clearly separated in contrast to the other plots. The low balanced accuracy in the type D test also indicates that the sliced Wasserstein k-means clustering algorithm struggles to seperate the bear and moon regimes while it captures well the bull trend. Comparing type C and D, regime detections we can conclude that the sliced Wasserstein algorithm struggles with the moon shape structure and not necessarily with the number of regimes. Indeed, we observe that for the three regimes setting in type C generated by GBMs the regime detection is still very accurate.

4.4.4 Discussion on higher dimensions

While the sliced Wasserstein k-means clustering algorithm with UnifOrtho performs consistently well on synthetic data type A, B and C, we observe that the performance of the algorithm improves for higher dimensions ($d = 100$) for type D data. Looking at the results in the appendix (table 8), we observe that for $d = 100$ the maximum total accuracy for type D data is always above 87%, which is not the case for lower dimensions. This echoes to subsection 3.4 in which we have observed qualitatively how the variance of the UnifOrtho results reduces as one increases the dimensions.

4.5 Summary

In this chapter, we have presented the results of the sliced Wasserstein k-means clustering on different types of synthetic data. The synthetic data was generated mostly using Geometric Brownian Motion (GBM) with different parameters to simulate different market regimes. We have shown that the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections is able to accurately detect regimes in synthetic data type A, B and C with very high accuracy and balanced accuracy across different hyperparameter settings as well as for different dimensionality (see Appendix). However, the algorithm struggles with more complex patterns such as with synthetic data type D, which combines GBM data with a moon-shape structure. Finally, we observe that the sliced Wasserstein k-means clustering with UnifOrtho performs globally better than the version given in Luan and Hamp (2025) [11] with pre-defined projections, particularly for higher dimensions and across different hyperparameter settings.

Chapter 5

Experiment 2: Sliced Wasserstein on real-world data sets.

This chapter applies the sliced Wasserstein k-means algorithm to real-world data. We first assess the quality of the detected regimes quantitatively, using a look-ahead trading strategy on three regional equity indices (USA, Europe and Asia), whose strong performance indicates that the algorithm separates regimes effectively. We then construct predictive signals from the cluster centroids based on the implied probability of a regime switch, and test them on the same data. The chapter concludes that these signals do not outperform long-only benchmarks, and that tuning the trading strategies does not reliably improve their performance.

5.1 Introduction

As synthetic data and particularly Geometric Brownian Motions (GBM) do not accurately reflect real world financial markets, this thesis also focuses on the application of the sliced Wasserstein k-means algorithm on equity index data sets across the USA, Europe and Asia. Moreover, regime detection literature overwhelmingly stop at labels and rarely backtest because the mathematical framework does not allow it. In the case of the sliced Wasserstein algorithm we have shown in section 3.5 that it is possible and we will therefore test it in a real-world environment. Additionally, the cross-regional comparison we set up in this experiment allows this research to truly understand on what market data the algorithm adapts and performs the best. The fundamental data set is often an overlooked aspect in studying models' efficiency which is addressed by this study. The focus of this chapter is therefore to compar and look at the performance of the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections across different regional clusters of Equity Indices (USA, Europe, Asia). To do so, this section is structured as follows. First, we will test wether the clustering results separate economically meaningful regimes using a look-ahead strategy. The focus will then shift to deploy the regime swith probability strategy and the clustering algorithm within realistic trading strategies on market data. Finally, we will examine how tuning the non-look-ahead trading strategies affects their performance on test data.

5.2 Background

This experiment consists of creating trading strategies that are based on the regime detection of the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections. In Luan and Hamp (2025) [11], authors suggested the use of the sliced Wasserstein k-means clusters to infer implied probability of regime switches which we developped in section 3.5. This in turn can be used to create trading strategies. Additionally, backtesting the latter is highly practical to test the accuracy of the above implied probability approach.

5.2.1 Hyperparameters

The parameters used for the developped trading strategy and regime detection for the first and second sub-experiment are given in table 5.1 below:

Symbol	Code variable	Value	Role
slicedWasserstein k-means (regime detection)			
h_1	`h1`	3	Lifting window length (support points per distribution)
h_2	`h2`	1	Lifting stride between windows
L	`L`	20	Number of projection directions
N_S	`N_S`	40	Random k-means restarts (best MCCD kept)
ε	`epsilon`	10^{-6}	Centroid-movement tolerance
K	`K`	2^*	Number of regimes / clusters
General backtest & signal aggregation (all strategies)			
C_0	`initial_capital`	100	Initial capital
w	`_window_size`	20	Rolling regime-estimation window ($\approx$1 month)
ℓ	`_majority_lookback`	20	Look-back for EW majority vote
λ	`_half_life`	5	Half-life of majority-vote weights
Implied-probability signal (Hysteresis / continuous / conviction)			
τ	`_tau`	adaptive	Softmax temp. of distance likelihood (`None`$\Rightarrow \frac{1}{2}$ mean inter-centroid dist.)
g	`_use_gradient`	True	Blend regime-momentum (gradient) term?
α	`_grad_weight`	$0.7^\dagger$	Gradient-term weight in P_{switch}
ω	`_lookback`	20	Look-back for OLS slope of distance-to-centroid
τ_{grad}	`_tau_gradient`	adaptive	Softmax temp. of gradient signal (`None`$\Rightarrow \frac{1}{2}$ std)
Hysteresis variant			
α_e	`_entry_threshold`	0.35	P_{switch} to enter counter-regime position
α_h	`_hold_threshold`	0.21	P_{switch} to restore regime-aligned position
Continuous variant			
δ	`dead_zone`	$0.10^\ddagger$	Dead-zone on $\pi_{\text{bull}} - \pi_{\text{bear}}$ (signal $\to 0$)
Conviction variant			
–	(slope)	$1.5^\ddagger$	Conviction decay $c = 1 - 1.5\,p_{\text{switch}}$

Table 5.1: Hyperparameters for the regime-detection and trading pipeline. * fixed by the two-regime setup, not a free parameter here. † `_grad_weight` is set but *inactive* while `_use_gradient=False`. ‡ hard-coded constant in the strategy code rather than a passed argument.

The above hyperparametrers are set as fixed for the look-ahead trading strategy, and all the other trading strategy implemented in this thesis. Interestingly, we do not provide a high h_1 value as suggested in the synthetic data test. This is because, in this experiment, we want to look a higher granularity of data in order to develop our trading strategies. Indeed, instead of creating our predictions on empirical distributions with high number of atoms as in the synthetic data test, we provide a refined segmentation of the time series with empirical distributions with a lower number of atoms. The goal is then to produce a regime prediction using an exponentially weighted average of the regime predictions on smaller intervals. Additionally, we observe that the `_window_size` is equal to 20 days, which means that we re-compute order size/direction every 20 business days ($\approx$ 1 month).

For the performance metrics we will be using the traditional Sharpe ratio as well as the Hit ratio defined below:

$$\text{Hit Ratio} := \frac{\text{Number of profitable trades}}{\text{Total number of trades}} \tag{5.1}$$

We use the Hit ratio to evaluate the effectiveness of the trading strategies. A higher Hit ratio indicates a higher proportion of profitable trades, which is desirable for a successful trading strategy.

We will also consider the Win/loss ratio, which is defined as the ratio of the average profit of winning trades over the average loss of losing trades.

$$\text{Win/Loss Ratio} := \frac{\text{Average profit of winning trades}}{|\text{Average loss of losing trades}|} \tag{5.2}$$

The Win/loss ratio provides insight into the risk-reward profile of the trading strategy. When this ratio exceeds 1, it indicates that the average profit from winning trades is greater than the average loss from losing trades.

Finally, we need to introduce the metric filter in our study. This metric filter is used on real world data sets to correctly assign labels 0 to bear regimes and 1s to bullish regimes. The issue arises from the fact that the sliced Wasserstein k-means clustering algorithm is only capable of separating regimes into different clustering labels, without automatically inferring the nature of the distinct clusters. However, looking at the characteristics of the different clusters and their corresponding centroids, we can infer the nature of the regimes and assign the labels accordingly. This is why, we introduce a `choose_label` function that sorts the labels based on either CVaR, Skewness or the MeanVar ratio of the returns in each cluster.

5.3 Data Set

For the experiments we will consider the following regional clusters of Equity Indices for our analysis.

- **America:** S&P 500, NASDAQ, Dow Jones (US Equity Cluster) (daily data)
- **Europe:** CAC 40, DAX40, FTSE 100 (European Equity Cluster) (daily data)
- **Asia:** Nikkei 225, Hang Seng, Kospi 200 (Asian Equity Cluster) (daily data)

For the sake of having statistically significant results, we consider 30 years of training data (1990- 2020) and 5 years of testing data (2020- 2025). We will use the training data for our preliminary analysis in the first subparts of the experiment, and to tuned our trading models. We will then only use the American testing data (2020 - 2025) for testing the tuned trading strategies.

Table 5.2 synthesizes the data sets used in this thesis for the real-world data set analysis.

Table 5.2: Summary of the data sets used for the analysis

Data Set	Equity Indices Studies	Number of Rows	Period Covered
America (Train)	S&P 500, NASDAQ, Dow Jones	7559	02/01/1990 - 31/12/2019
America (Test)	S&P 500, NASDAQ, Dow Jones	1258	02/01/2020 - 31/12/2024
Europe (Train)	CAC 40, DAX40, FTSE 100	7427	02/01/1990 - 30/12/2019
Asia (Train)	Nikkei 225, Hang Seng, Kospi 200	6773	04/01/1990 - 30/12/2019

5.4 Results and Discussion

This experiment will be split into three parts. First, we will look at an objective performance analysis of the regime detection on the real world data sets by using a long/flat look-ahead trading strategy with signal generated using the current regime estimation. This will then alllow us to create real-world trading strategies with multiple variations for signal generation. The performance will then be compared to a long-only benchmark and a short-only benchmark. The tuning of the hyperparamets in the above experiments will be chosen according to the table 5.1. Finally, we will look at the performance of the different trading strategies under different hyperparameters to understand the impact of specific settings on the trading performance.

5.4.1 Sliced Wasserstein Regime Detection with Look-Ahead

In this subsection we want to look at the performance of the regime detection algorithm using a trading strategy with look-ahead bias to understand whether the regime detection is able to detect the regimes on a day to day basis, without considering predictive signals. We can visually observe in the figure 5.1, 5.2 and 5.3 that the sliced Wasserstein regime detection algorithm labels well market periods for the three regions with downturns and upturns. Green represents the bullish regimes, while red represents bearish regimes. The plots below were generated using the parameters in table 5.1 for the sliced Wasserstein regime detection algorithm.

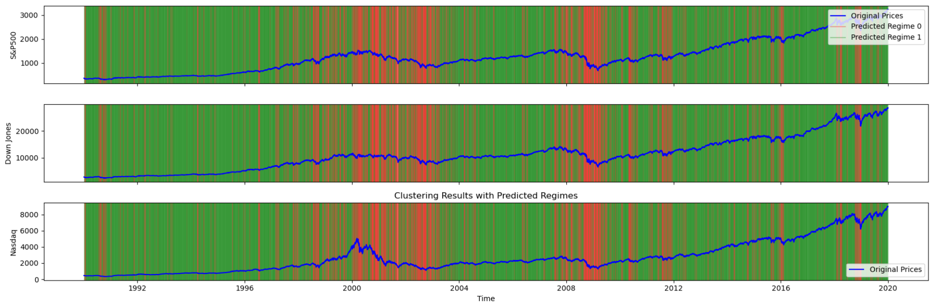


Figure 5.1: Regime labels output by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections on the US equity indices (S&P 500, NASDAQ, Dow Jones) time series. Parameters $h_1 = 3$, $h_2 = 1$, $L = 20$, $N_S = 40$.

Figure 5.2: Regime labels output by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections on the European equity indices (CAC 40, FTSE 100, DAX) time series. Parameters $h_1 = 3$, $h_2 = 1$, $L = 20$, $N_S = 40$.

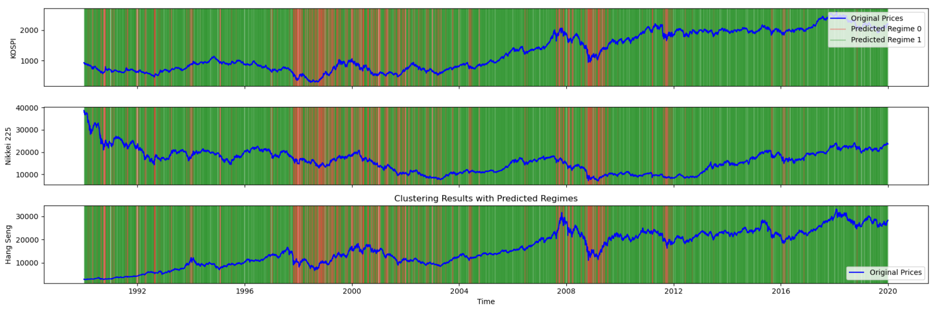


Figure 5.3: Regime labels output by the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections on the Asian equity indices (Nikkei 225, Hang Seng, Kospi 200) time series. Parameters $h_1 = 3$, $h_2 = 1$, $L = 20$, $N_S = 40$.

The objective of this experiment is to transform the above qualitative observation into a quantitative benchmark of the regime detection performance on a real-world data set in which no "true-label" is available as market regimes are subject to interpretation. The pseudo code for the look-ahead trading strategy for this experiment is available in the appendix (2). We test our look-ahead trading strategy on the 3 regional inverse-vol weighted index portfolios. This is done with an inital capital of 100, over the window 1990-2020 and using the hyperparameters in table 5.1. We present the performance of the look-ahead strategies on the three regional markets in figure 5.4, 5.5 and 5.6. Additionally, a summary with key performance metrics is also provided in Table 5.3.

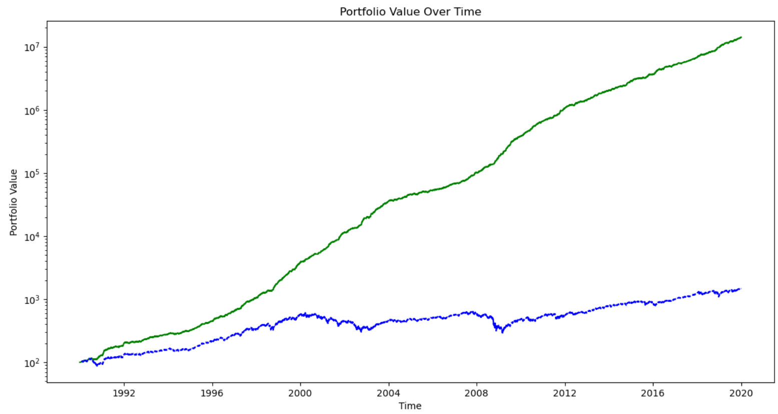


Figure 5.4: Look ahead trading strategy on the inverse vol weighted portfolio of US equity indices (S&P 500, NASDAQ, Dow Jones) with initial capital of 100. The trading strategy is based on the regime detection using the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections. The trading strategy is a long/flat strategy based on the current regime detection signal.

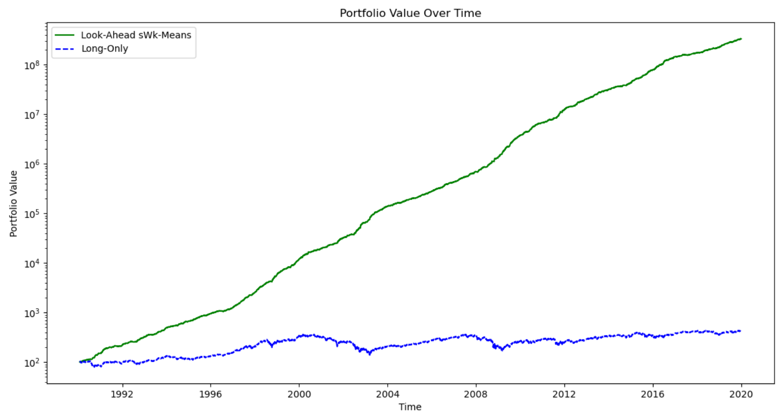


Figure 5.5: Look ahead trading strategy on the inverse vol weighted portfolio of European equity indices with initial capital of 100. The trading strategy is based on the regime detection using the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections. The trading strategy is a long/flat strategy based on the current regime detection signal.

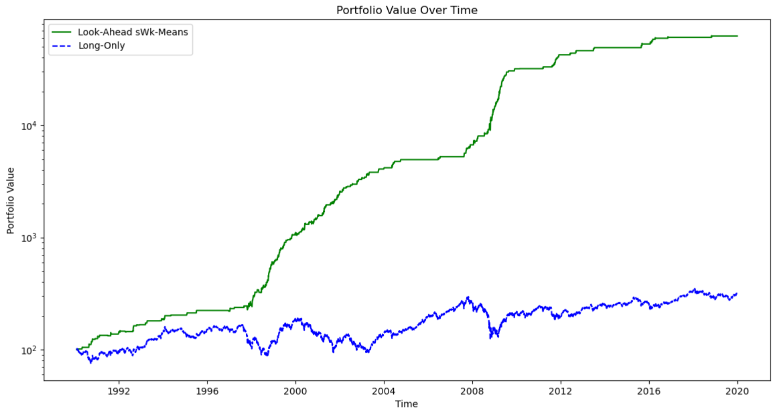


Figure 5.6: Look ahead trading strategy on the inverse vol weighted portfolio of Asian equity indices with initial capital of 100. The trading strategy is based on the regime detection using the sliced Wasserstein k-means clustering algorithm with UnifOrtho projections. The trading strategy is a long/flat strategy based on the current regime detection signal.

Table 5.3: Cross-Regional Portfolio Performance Summary

Region	Strategy Config	Metric Filter	Hit Ratio	Win/Loss	Ann. Sharpe	Ann. Return
America	Look-Ahead UnifOrtho	CVaR ($K = 2$)	0.5987	1.4994	3.9853	52.24%
Europe	Look-Ahead UnifOrtho	CVaR ($K = 2$)	0.6066	1.6225	4.3411	64.92%
Asia	Look-Ahead UnifOrtho	Skewness ($K = 2$)	0.6775	1.6299	1.99	23.95%

Note: Initial capital $I_0 = 100$.

- *US* reached a final value of 29,911,844.77 (Cumulative PnL: 29,911,744.77).
- *Europe* reached a final value of 329,792,991.92 (Cumulative PnL: 329,792,891.93).
- *ASIA* reached a final value of 62,503.61 (Cumulative PnL: 62,403.61).

We observe that in the above three plots, the look-ahead trading algorithm generated from the labels of the sliced Wasserstein k-means clustering algorithm performs well on the different equity indices. In particular, the Win/loss ratio is above 1.4 for all three regions, which indicates that the average profit from winning trades is significantly higher than the average loss from losing trades.

This give us a strong indication that under real empirical distributions the sliced Wasserstein k-means algorithm performs well. However, we observe that while performance is very good for the US and European markets it excels relatively less in Asian markets. This is particularly noticeable as US and European performance with a look-ahead trading strategy have an annualised Sharpe ratio of 4 or more, while the Asian look-ahead only reaches timidly 2.0 annualised Sharpe ratio. The performance gap can also be seen as the average annual return for the strategies in US and Europe go beyond 50% while the Asian strategy only reaches 23.95% average annual return. Qualitatively, we also observe on figures 5.1 and 5.2 that the sliced Wasserstein k-means clustering detects better bear regimes than the Asian counterparts, which translated in the performance gap in the look-ahead trading strategy. Interestingly, on the look-ahead trading strategy for the Asian equity indices we observe catastrophic results when using the metric filter based on CVaR. It seems that the CVaR captures bear regimes badly and misclassifies them as bullish regimes, which leads to a very poor performance of the trading strategy.

5.4.2 Trading strategies using Sliced Wasserstein

Having quantitatively and qualitatively established that the sliced Wasserstein k-means clustering algorithm performs well on real-world data sets, we now want to look at the performance of the sliced Wasserstein k-means algorithm for generating predictive signals for trading strategies. In this experience multiple trading strategies – listed below – are therefore considered.

- **SW k-means Long/Short** based on exponential weighted majority of the last l regime labels detected by the sliced Wasserstein algorithm with l the window size.
- **Implied strategy** based on Implied probability of regime switch as defined in 3.25 and the pseudo code in 4. This strategy is split into three variants.

  Implied Hysteresis : Using threshold on the switch probability to enter long or short positions and to exit the position.

$$\text{Enter Position} = \begin{cases} \text{Long} & \text{if } P_{\text{switch}} > \alpha_e \\ \text{Short} & \text{if } P_{\text{switch}} < 1 - \alpha_e \end{cases} \tag{5.3}$$

  Implied Continuous : Using the raw difference in prior probability to enter long/short positions.

$$\text{Position Size} = \pi_{\text{bull}} - \pi_{\text{bear}} \tag{5.4}$$

  Implied Conviction : Adjusting the position size based on the significance of the switch probability.

$$\text{Position Size} = 1 - 1.5 \cdot P_{\text{switch}} \tag{5.5}$$

- **Ensemble Methods**: combining the above four strategies using a weighted average of the signals or an adaptative weighting scheme to generate the final trading signal.

For curious readers all the above trading strategies are described in more detail in their respective pseudo codes in the appendix (3, 4). In our experimental settup we will be using the hyperparameters in table 5.1 for the regime detection and trading strategies. The trading strategies will be tested on a portfolio of inverse-volatility weighted equity indices for the three regions (US, Europe, Asia) and compared to a long-only benchmark as well as to a short-only benchmark. The initial capital is 100 and the strategies are tested on the training data from 1990 to 2020.

Additionally, we backtest two different ensemble strategies with different weighting schemes. One ensemble strategy is constructed with adaptative weighting based on the performance of the different trading strategies. The other ensemble strategy is deployed with equal weights for all the strategies on the final signal.

Below, in figure 5.7, 5.8 and 5.9, we plot the performance of the portfolio value applied to the different strategies for the three regions, as well as with the associated swtich probability as defined in 3.25 over time. We also give a summary of the performance of each strategy in tables 5.4, 5.5 and 5.6.

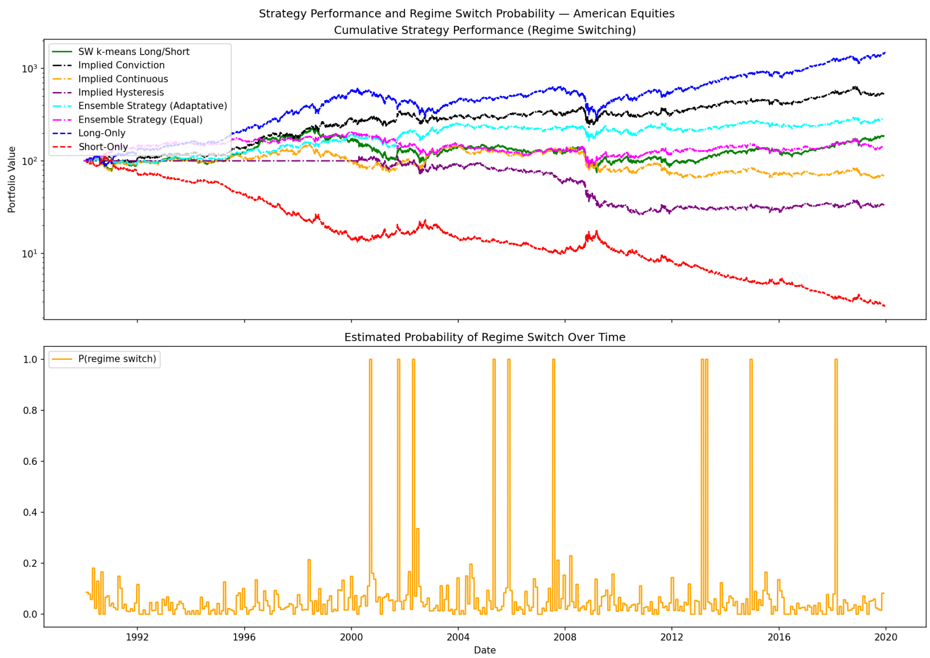


Figure 5.7: Trading strategies backtest on the inverse vol weighted portfolio of American equity indices (1990-2020) with initial capital of 100.

Table 5.4: Compact summary: Sharpe ratio and annualised return of each strategy relative to the long-only benchmark, US equity cluster, 1990–2020.

Strategy	**Ann. Sharpe**	**Ann. Ret. (%)**	Hit ratio	Win/Loss ratio
Long only (benchmark)	0.5992	9.41	0.5434	0.9392
Implied Conviction	0.4262	5.79	0.5285	0.9658
Ensemble Adaptative	0.3150	3.54	0.5205	0.9791
SW k-means Long/Short	0.2065	2.11	0.5224	0.9500
Ensemble Equal	0.1527	1.17	0.5157	0.9681
Implied Continuous	0.0044	−1.18	0.5109	0.9567
Implied Hysteresis	−0.1561	−3.54	0.4981	0.9722
Short only	−0.5992	−11.39	0.4565	1.0648

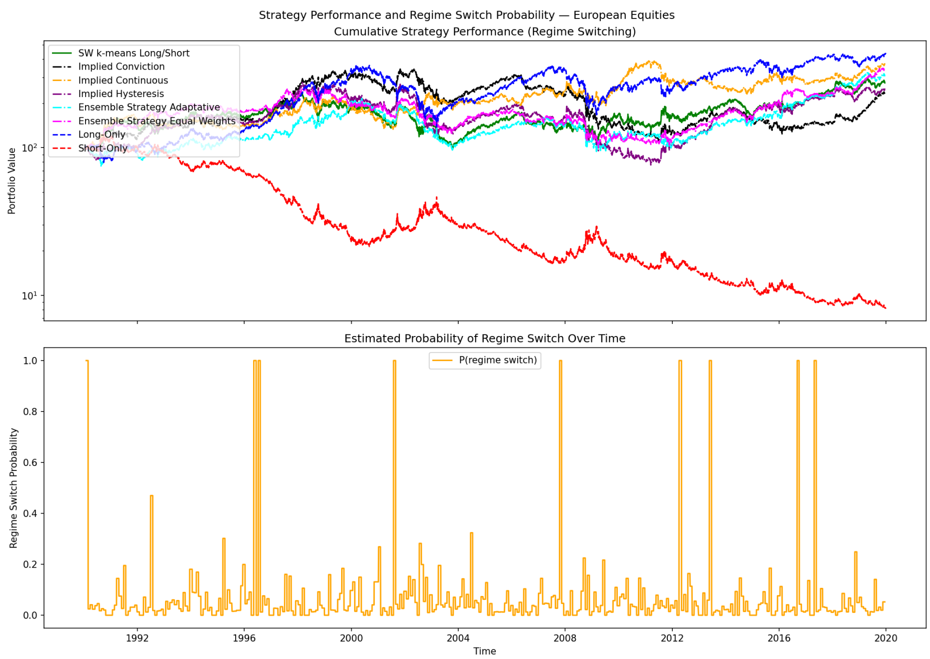


Figure 5.8: Trading strategies on the inverse vol weighted portfolio of European equity indices (1990-2020) with initial capital of 100.

Table 5.5: Compact summary: Sharpe ratio and annualised return of each strategy relative to the long-only benchmark, European equity cluster (CAC 40, DAX 40, FTSE 100), 1990–2020.

Strategy	**Ann.Sharpe**	**Ann. Ret. (%)**	Hit ratio	Win/Loss ratio
Long only (benchmark)	0.3560	5.01	0.5265	0.9591
Ensemble Equal	0.3388	4.26	0.5144	1.0047
Implied Continuous	0.3500	4.50	0.5057	0.9939
SW k-means Long/Short	0.2750	3.44	0.5102	1.0091
Implied Hysteresis	0.2592	3.14	0.5133	0.9938
Implied Conviction	0.2471	2.81	0.5117	0.9980
Ensemble Adaptative	0.2378	2.55	0.5162	0.9795
Short only	−0.3560	−8.00	0.4734	1.0426

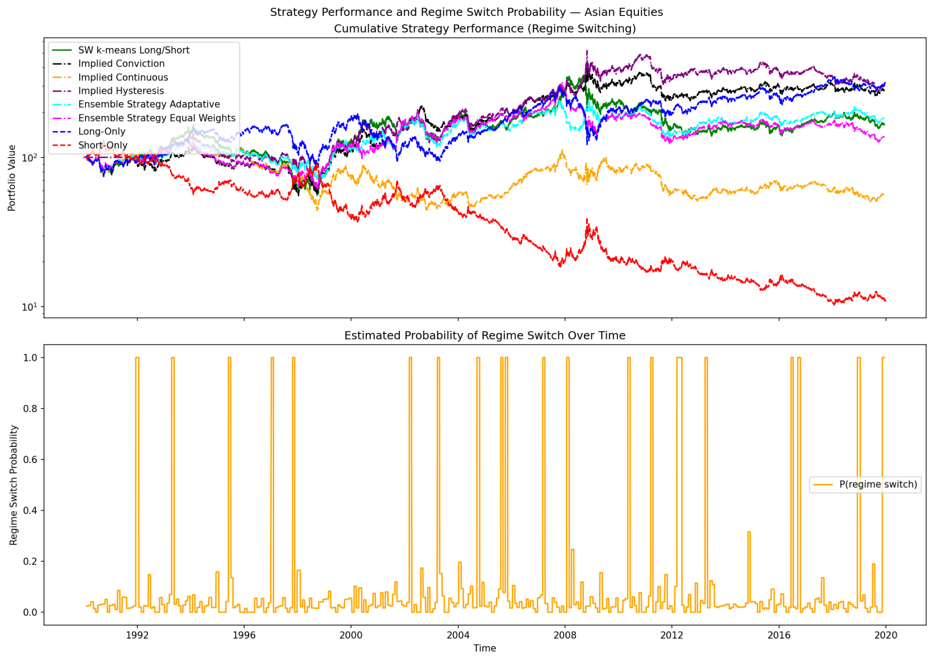


Figure 5.9: Trading strategies on the inverse vol weighted portfolio of Asian equity indices (1990-2020) with initial capital of 100.

Table 5.6: Compact summary: Sharpe ratio and annualised return of each strategy relative to the long-only benchmark, Asian equity cluster (Nikkei 225, Hang Seng, Kospi 200), 1990–2020.

Strategy	**Ann. Sharpe**	**Ann. Ret. (%)**	Hit ratio	Win/Loss ratio
Long only (benchmark)	0.2971	3.89	0.5238	0.9624
Implied Hysteresis	0.2926	3.75	0.5082	1.0258
Implied Conviction	0.2845	3.54	0.5122	1.0062
Ensemble Adaptative	0.2062	2.03	0.5072	1.0119
SW k-means Long/Short	0.1852	1.73	0.5075	1.0059
Ensemble Equal	0.1457	1.05	0.5100	0.9868
Implied Continuous	−0.0283	−1.90	0.5027	0.9775
Short only	−0.2971	−7.11	0.4760	1.0391

We observe that for all the three regions, the trading strategies based on the sliced Wasserstein k-means clustering algorithm underperform the long-only benchmark on the 30 year long benchmark. The performance of the best performing strategy in the US region has an annualised return ($\approx 5\%$) which is almost half of the annualised return of the US long-only benchmark ($\approx 9\%$). However, this performance gap is far tigher in the european and asian regions.

Looking at the probability switch plot below the portfolio value plot, we observe that the signal is not consistently reliable. While, in many cases, we see clear regime switch probability spikes slightly before a clear market downturn, the model also lags in flagging recovery periods as we can observe this during the recovery of the 2008 financial crisis. This is particularly true for european and american equities indices. Moreover, this lagg in recovery periods make the strategies lose on the long-term against a long-only benchmark as we see in the Asia region from 2008 to 2020.

Moreover, even though the implied probability signal flaggs many market downturns before their happen, this advantage is not entirely leveraged by the different strategies as their mostly take the shock during bear regimes. Therefore, one reaches the conclusion that the forecasts generated by the implied probability of regime switch are not entirely reliable even though their flag many early signs of bullish/bearish switch. The main issue is also being able to efficiently trade this probability forecast, which is currently not the case.

Finally, we observe that in all three regions, the trading strategies have consistently better Win/Loss Ratio than the long only benchpmark but never exceed its Hit ratio metric at around $(0.52 - 0.54\%)$. This indicates that in order to outperform the market the trading strategies must be able to get a higher number of positive trades instead on only having average higher profits.

5.4.3 Hyperparameter tuning

In this section we want to expliore the influence of the hyperparameters as shown in table 5.1 on the performance of the trading strategies (SW k-means Long/Short, Implied probability, Ensemble methods) and the regime detection algorithm (sliced Wasserstein k-means clustering with UnifOrtho projections).

Tuning the gradient weight for implied probability methods In this research, multiple values for the gradient weight α were tested on the equally weighted ensemble method to observe how the overall performance of equally weighted implied probability strategies (Hysteresis, Continuous, Conviction) is affected by the gradient weight α as expressed in equation 3.25. The values below are tested on the American equity index data set from 1990 to 2020 with an initial capital of 100. Other hyperparameters except α are set according to Table 5.1.

$$\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\} \tag{5.6}$$

In Figure 5.10 we have plotted the performance of the ensemble method as the combination of Hit ratio and Win/loss ratio against the gradient weight α.

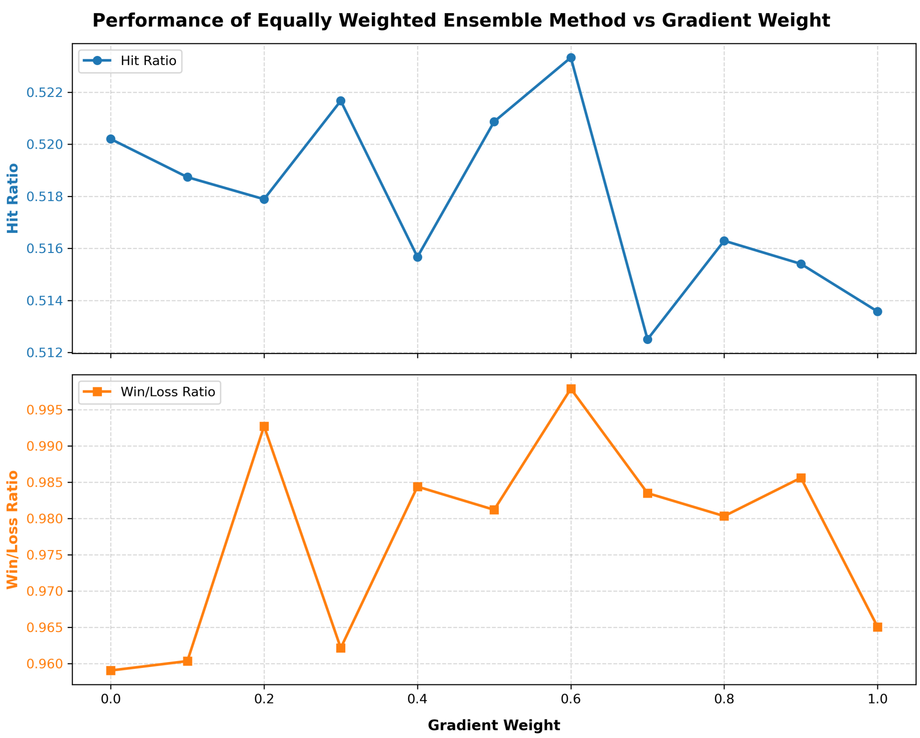


Figure 5.10: Performance of the equally weighted ensemble method as a function of the gradient weight α on the American equity data set from 1990 to 2020 with an initial capital of 100. The performance is measured as the combination of Hit ratio and Win/loss ratio.

We observe in the above figure that the best performance is achieve for $\alpha = 0.6$ with a peak in Hit ratio and Win/loss ratio for this parametrization. This is interesting as this clearly suggest that the momentum has more positive influence on the trading signals in the implied probability strategy then the knowledge of the current regime. This result is counter-intuitive as one would have expected a lower value of α, describing the persistent nature of regimes in financial markets. However, we need to point out that the results plotted above are highly variable depending on the random initial centroids of the sliced Wasserstein algorithm. Hence, this study is not an proof of the importance of momentum in regimes and would require a dedicated research on the statistical significance of the above plots.

Tuning the temperature parameter τ and τ_{grad} for the implied probability strategy . Additionally, we performed a grid search for τ for the switch probability in 3.17 in combination with τ_{grad} from equation 3.24. The objective is to understand wether those parameters influence the accuracy of the implied probability prediction signals. The optimal values for τ and τ_{grad} are determined based on the highest Hit ratio and Win/loss ratio achieved during the backtesting period (1990-2020) on american equity index data. When looking at the heatmaps we denote τ on the x-axis and τ_{grad} on the y-axis. "Nan" values denote τ and τ_{grad} combinations that are computed as the mean of the centroid distances and the mean of the gradient signals as follows:

$$\tau_{\text{nan}} := \frac{1}{N(N-1)} \sum_{k<k'} \|c_k - c_{k'}\|_2, \quad \tau_{\text{grad,nan}} := \frac{1}{2N} \sum_{k=1}^{K} (\beta_k - \bar{\beta}_k)^2 \tag{5.7}$$

with c_k being the centroid of the k-th cluster and β_k being the gradient signals as computed in equation 3.23. $\bar{\beta}_k$ denotes the average gradient signal for the k-th cluster with $k \in \{1, \dots, K\}$.

In Figures 5.11 and 5.12 we have plotted the results of the grid search for the Hit ratio and the Win/loss metric respectively.

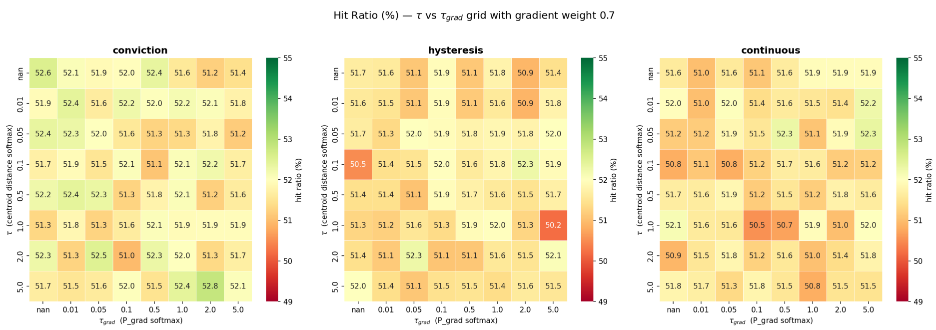


Figure 5.11: Performance in terms of Hit ratio of the implied probability strategy as a function of the temperature parameters τ and τ_{grad} on the American equity data set from 1990 to 2020 with an initial capital of 100.

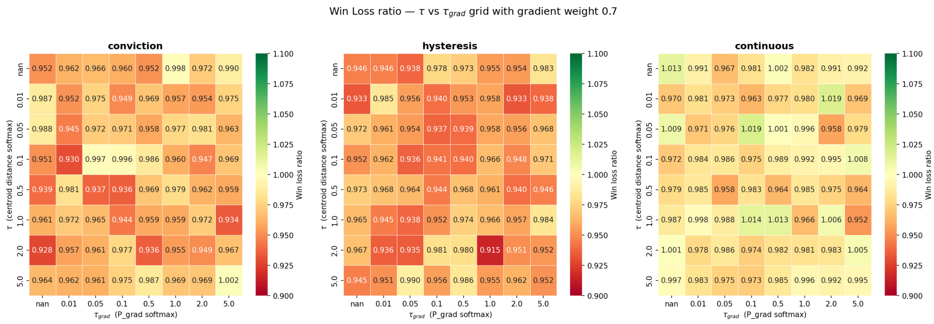


Figure 5.12: Performance in terms of Win/Loss ratio of the implied probability strategy as a function of the temperature parameters τ and τ_{grad} on the American equity data set from 1990 to 2020 with an initial capital of 100.

In the above heat maps we cannot make any clear conclusion on the impact of the temperature parameters τ and τ_{grad} on the performance of the implied probability strategies. In fact there seems to be no significant trend in the Win/loss ratio and Hit ratio as one changes those parameters.

Tuning the half-life parameter for the exponentially weighted majority voting scheme. In this subsection we we want to explore the influence of the half-life parameter λ on the trading strategies using Hit ratio and Win/loss ratio as performance indicators. We want to look at the evolution of the performance for all the developped strategies :

- the SW k-means long/short strategy and
- the implied probability strategy with Hysteresis, continuous and conviction variations.
- the ensemble method with equally weighted signals.

In addition the teste values for the half-life parameter λ are the following:

$$\lambda \in \{1.0, 3.0, 5.0, 10.0, 20.0\} \tag{5.8}$$

Again we used the parameters in Table 5.1 for the other hyperparameters and we used the American equity data set from 1990 to 2020 with an initial capital of 100. We present the results in Figure 5.13.

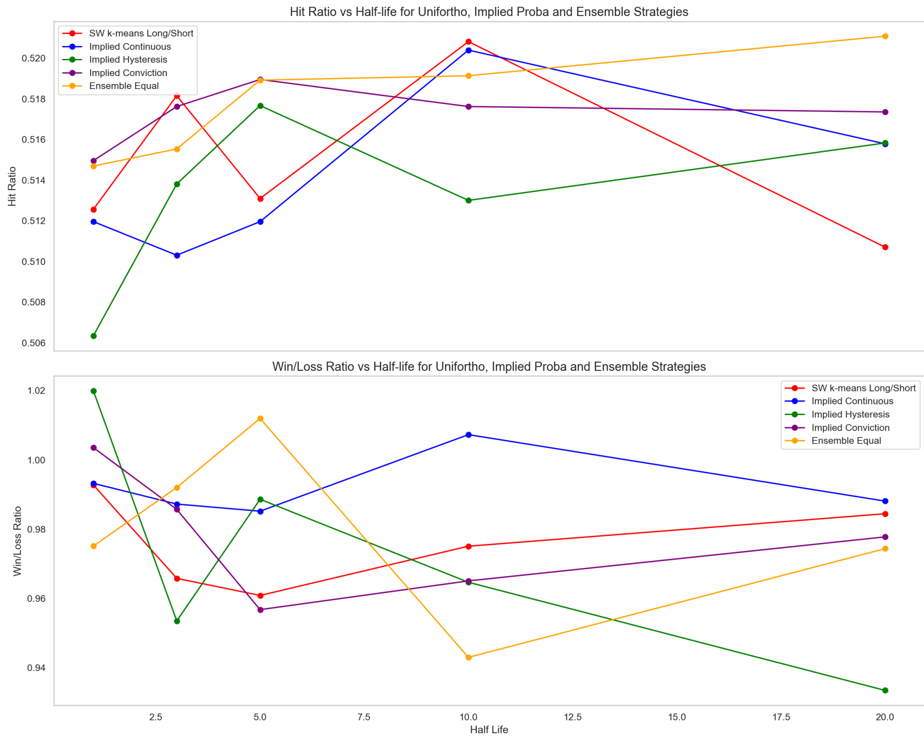


Figure 5.13: Performance of the equally weighted ensemble method as a function of the half-life parameter λ on the American equity data set from 1990 to 2020 with an initial capital of 100. The performance is measured as the combination of Hit ratio and Win/loss ratio.

We observe that no clear half-life parameter λ stands out in our tests that would be optimal for all strategies. The variability of the Hit ratio and Win/loss ratio is very high for the different parameters and no clear trend can be observed. However, one can observe that for the SW k-means Long/Short strategy and the implied continuous strategy one observes a spike in Hit ratio at a half-life $\lambda = 10$. On the other hand, for the implied Hysteresis and ensemble equal the spike in Hit ratio and Win/loss ratio is located at $\lambda = 5$.

Tuning the trading threshold for the Hysteresis strategy. In this research we are also interested in exploring how to optimally parametrize the entry and hold threshold of the Hysteresis strategy. For this purpose, we performed a grid search on the hold and entry threshold by looking at the performance of the Hysteresis strategy in terms of Hit ratio and Win/loss ratio. The tested values for the hold and entry threshold are the following:

- Hold threshold: $h \in \{0.2, 0.3, 0.35, 0.4\}$
- Entry threshold: $e \in \{0.25, 0.3, 0.35, 0.4, 0.45, 0.5\}$

Those chosen parameters are based on previous testing. Again we used the parameters in table 5.1 for the other hyperparameters and we used the American equity data set from 1990 to 2020 with an initial capital of 100. The testing values above also are based on the switch probability range observed given the parameters in table 5.1. We present the results in Figures 5.14 and 5.15 for the Hit ratio and Win/loss ratio metrics respectively.

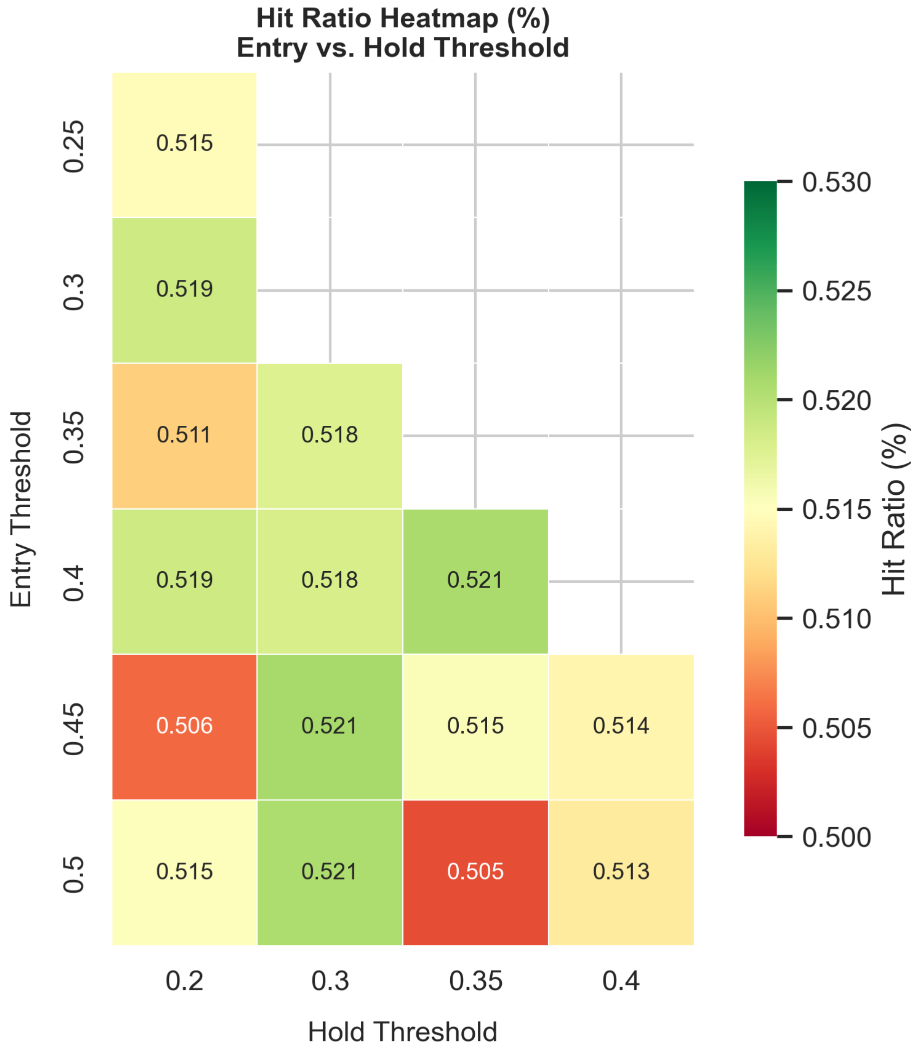


Figure 5.14: Hit Ratio of the Hysteresis strategy as a function of the hold and entry threshold on the American equity data set from 1990 to 2020 with an initial capital of 100.

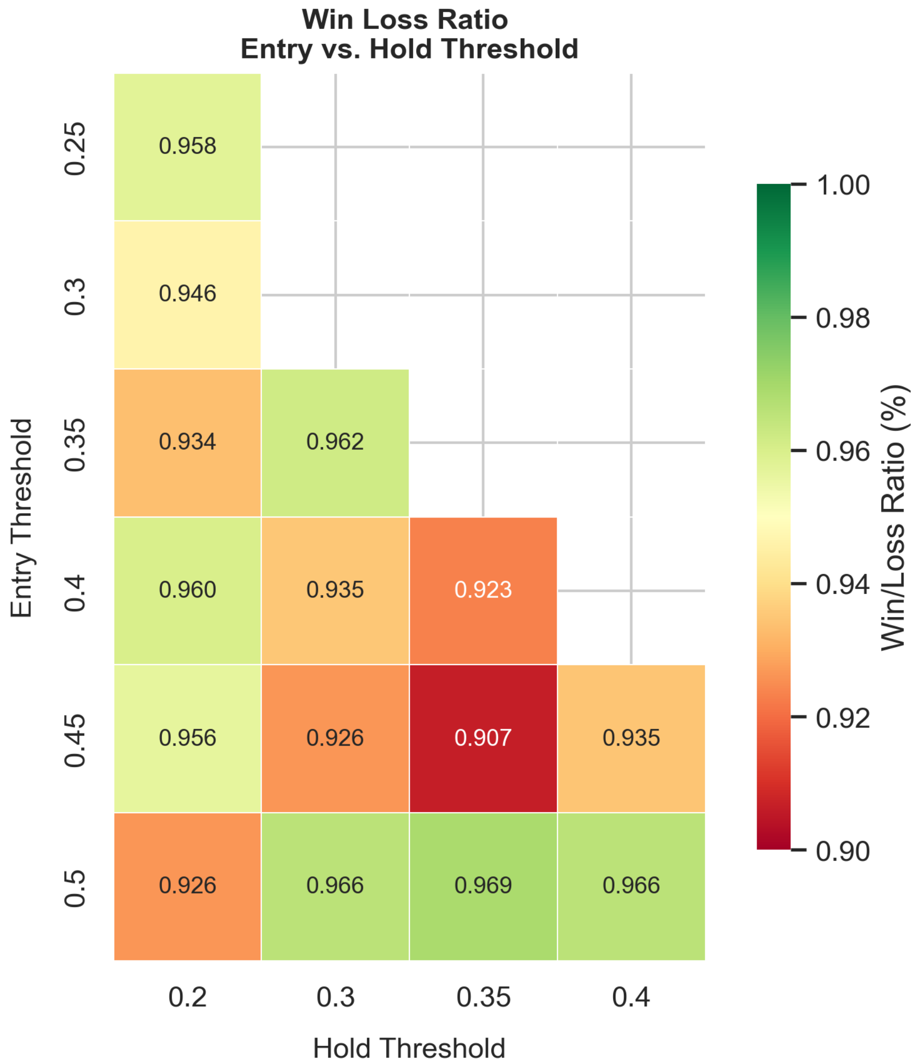


Figure 5.15: Win/Loss Ratio of the Hysteresis strategy as a function of the hold and entry threshold on the American equity data set from 1990 to 2020 with an initial capital of 100.

We observe that the tuning of entry and hold thresholds that are consistently the highest across the Hit ratio and Win/loss ratio given the hyperparameters in table 5.1 is the combination of $(entry, hold) \in \{(0.4, 0.20), (0.5, 0.3)\}$ with Hit ratio and Win/loss ratios couples that are respectively $(0.519, 0.960)$ and $(0.521, 0.966)$. Looking at other combinations gives acceptable Hit ratios and catastrophic Win/loss ratio which shows instability. One has to point out that the Win/Loss ratios in the Hysteresis strategies are not particularly high as they remain consistently below 1.0. This indicates that the strategies lose more than their gain on average. Again, we need to point out that the results plotted above are highly variable depending on the random initial centroids of the sliced Wasserstein algorithm. Hence, this study is not a proof of the importance of threshold tuning in regimes and would require a dedicated research on the statistical significance of the above plots.

5.4.4 Simulation on test data (post-2020)

After having looked at the influence of the hyperparameters on the performance of the different trading strategies, we test a tuned version of the equally weighted ensemble method which we call "Ensemble Tuned" strategy on the test data set from 1st of January 2020 to 31st of December 2024 on the inverse-vol weighted portfolio of American Indices with an initial capital of 100 against a long-only benchmark. We test the equally weighted ensemble method with the tuned parameters from the previous section (i.e $\alpha = 0.6$ on $hold\_threshold = 0.3$ and $entry\_threshold = 0.5$). The other parameters are kept identical to the tuning in table 5.1.

Moreover, we want to look at the performance of the tuned equally weighted ensemble strategy in which one does not short when bear regime is predicted but instead goes to cash. We will call this more conservative approach the "Defensive" strategy.

Finally, we will compare the performance all these strategies to an equally weighted portfolio of long-only strategy and the ensemble method strategy which we call the "Overlay" strategy. The performances of the out of sample strategies are summarized in table 5.7 below and plotted in Figure 5.16.

Table 5.7: Out-of-sample performance of the equally weighted ensemble method with $\alpha = 0.6$ on the American equity data set from 1st of January 2020 to 31st of December 2024 with an initial capital of 100.

Strategy	**Sharpe**	**Ann. Return**	**Hit Ratio**	**Win/Loss Ratio**
Long Only	0.6694	12.82 %	0.5360	0.9834
Defensive	1.2005	11.82 %	0.5495	1.0277
Overlay	0.9280	11.02 %	0.5369	1.0420
Ensemble Tuned	0.5675	7.77 %	0.5283	1.0009

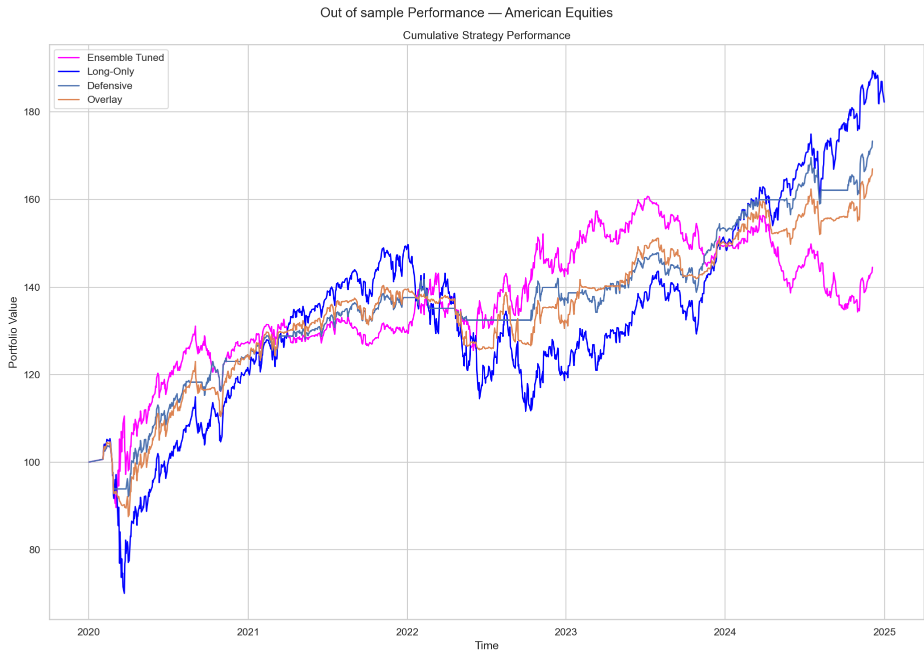


Figure 5.16: Out-of-sample performance of the equally weighted ensemble method with $\alpha = 0.6$ on the American equity data set from 1st of January 2020 to 31st of December 2024 with an initial capital of 100.

We observe that the Defensive strategy slightly underperforms the long-only benchmark on the overall 5-year. While its Sharpe ratio, Hit ratio and Win/Loss ratio is higher than the long-only benchmark, its annualized return is below the long-only benchmark. Comparing it to the Ensemble and Overlay strategies, we observe that, going flat during bear is highly beneficial to the overall performance. In fact, when looking at the Ensemble strategy, we observe that it performs better by shorting during bear markets but its late switch back to long when the regime changes signficantly hurts the performance and increases the volatility of the strategy. This is particularly visible when comparing Overlay and Defensive strategy. While they both have similar annuanlised return (11.02% and 11.82% respectively), the Defensive strategy has a much higher Sharpe ratio (0.9280 vs. 1.2005), indicating a better risk adjusted return.

5.5 Summary

We observe that while the look-ahead trading strategies based on the sliced Wasserstein clustering results overperforms the long-only benchmark, the real trading strategies using predictive signals underperform the long-only benchmark on all regional equity index portfolios. This is mainly due to the fact that the predictive signals are not accurate enough to generate profitable trading signals. Moreover, while the regime switching probability flags some bear regimes, the late switch back to long positions when the market has already recovered significantly hurts the performance of the trading strategies. Tuning the hyperparameters of the trading strategies mostly does not significantly have an effect on the performance of the trading strategies in terms of Hit ratio and Win/loss ratio. We observe nonetheless small trends notably for the gradient weight α in terms of Hit ratio and Win/loss ratio metrics. In particular, one conclusion from the tuning is that momentum seems to have more influence on the performance of the trading strategies than the knowledge of the current regime. Finally, the out-of-sample test on the American equity data set from 2020 to 2024 shows that the Defensive strategy parametrized using the tunig of the hyperparameters in the training data set (1990-2020) performs better than shorting during bear. This shows again that the recovery period after bear markets are ill-handled by the predictive signals.

Chapter 6

Experiment 3: Interpretability of clustering results

This chapter studies how macroeconomic indicators relate to the statistical output of the sliced Wasserstein k-means algorithm. We fit supervised learning models to macroeconomic data spanning 2006 to 2020, using the algorithm's regime labels over that period as targets, and analyse the resulting feature importances. The bear regimes flagged by the model coincide with recognised periods of macroeconomic stress, strengthening the interpretability of the sliced Wasserstein k-means approach.

6.1 Introduction

While it is of significant interest to practitioners to generate signals from regime detection algorithms, it is almost equally important in the industry to understand the output of a model. In particular, having an interpretation of the outputs of the sliced Wasserstein algorithm in terms of macroeconomic indicators can be of great interest for an enhanced understanding of the applicability of the latter regime detection method. We want to specifically understand whether the statistical and mathematical methods used to detect bear regimes coincide with real periods of economic downturns and market uncertainty. To do so, we first explain the SHAP analysis framework which will be used for conducting the experiment. A SHAP analysis will then be applied to linear regression, logistic regression and random forest classifier models for understanding feature importance during flagged bear regimes.

6.2 Background

6.2.1 SHAP Analysis

To perform this study, we analyse the feature importance of different macro-indicators during predicted bear regimes on the output of supervised learning models using the SHAP analysis toolbox introduced by Lundberg et al.(2017) [12]. In order to understand the SHAP methodology, we give the following notations. Let us denote by f the supervised learning model, and let $F = \{1,\dots,M\}$ be the set of features – in our case the macroeconomic indicators – and let $x = (x_1,\dots,x_M) \in \mathbb{R}^M$ be a specific input. For any subset of features $S \subseteq F$ we denote by x_S the values of the features in S and by $x_{\bar{S}}$ the values of the features not in S. We define the value function $v(S)$ as :

$$v(S) := \mathbb{E}\big[f(X) \mid X_S = x_S\big], \qquad S \subseteq F = \{1,\dots,M\}. \tag{6.1}$$

[12]

The SHAP value or Shapley value ϕ_i of the i-th feature is then defined as the average marginal contribution of the i-th feature to all possible subsets of features $S \subseteq F \setminus \{i\}$, weighted by the size of the subset S and the total number of features M:

$$\phi_i := \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,\big(M - |S| - 1\big)!}{M!} \Big[v\big(S \cup \{i\}\big) - v(S)\Big]. \tag{6.2}$$

In the following experiment we use the SHAP value because of its properties as described in [12]. Indeed, the Shapley value is the only feature attribution method which satisfies:

- **Local accuracy**: The shap value decomposes the individual prediction exactly:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{6.3}$$

- **Missingness**: A feature absent from on instance receives no attribution $\phi_i = 0$.
- **Consistency**: If a model changes so that the marginal contribution of a feature increases or stays the same regardless of other features, the attribution for that feature should not decrease.

Those properties intuitively describe what a feature contribution should satisfy and therefore make the SHAP value a perfect candidate for our analysis.

6.3 Data Set

In our experiment we use the regime detection algorithm on weekly equity data (S&P 500, NASDAQ, Dow Jones) on the period from the 6th of Jan 2006 to 20th of December 2019. The sliced Wasserstein regime detection algorithm is then applied with the parameters $h_1 = 7, h_2 = 3, L = 20, N_S = 100, \varepsilon = 10^{-6}, K = 2$. We use slightly higher h_1 values to account for the fact that we are not using the sliced Wasserstein for a trading strategy but for pure regime detection. However, yet again we do not use higher h_1 values as in the synthetic data experiment in Chapter 4 due to the low availability of data (higher h_1 values requiring longer study period). To help us in the interpretation of the regimes we consider the following list of macroeconomic indicators on the study period.

- VIX Index
- 10-year Treasury Yield
- 5-year Treasury Yield
- 30-year Treasury Yield
- Fed Funds Rate
- Manufacturing PMI
- Manufacturing Employment
- DXY Dollar Index
- Gold Spot Price
- Brent Crude Oil Spot Price

We give in Table 6.1 a brief summary table of the data sets considered for the experiment.

Table 6.1: Summary of the data sets used for the analysis.

Data Set	Frequency	Number of Rows	Period Covered
Index Data (S&P 500, NASDAQ, Dow Jones)	Weekly	729	06/01/2006 - 20/12/2019
Macroeconomic Indicators	Weekly	729	06/01/2006 - 20/12/2019

This list is used as feature inputs for the supervised learning methods below to fit onto the regime classification generated by the sliced Wasserstein k-means clustering algorithm.

- OLS Linear Regression (probabilistic target: Probability of being in bear regime)

- Logistic Regression (binary target : Bull/Bear)
- Random Forest Classifier (binary target: Bull/Bear)

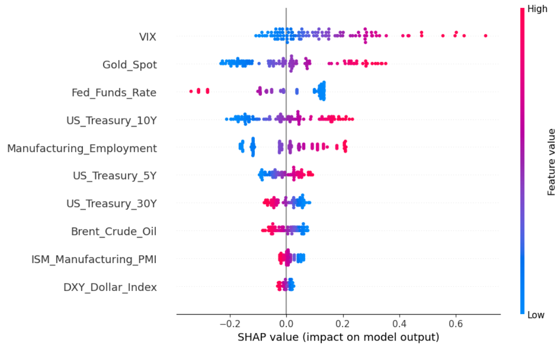
In the OLS Linear Regression, we use the probability of being in a bear regime as the target variable. This probability is computed using the regime labels generated by the sliced Wasserstein k-means method. Those labels are then transformed into a likelihood of being in a bear regime by using the softmax funtion introduced in equation 3.17. From this probabilistic target, we can then fit the OLS linear regression model to understand the relationship between the macroeconomic indicators and the probability of being in a bear regime. In the two other supervised learning methods (Logistic Regression and Random Forest Classifier), we use the binary regime labels generated by the sliced Wasserstein k-means method directly as the target variable. We encode the bear regime as 1 and the bull regime as 0.

6.4 Results and Discussion

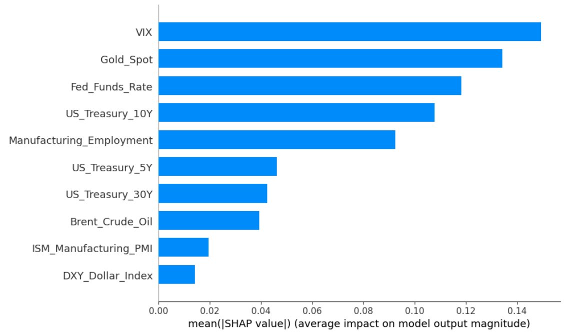
We observe the following accuracy score for the three supervised learning methdos on the training data :

- OLS Linear Regression ROC(AUC) score: 0.980
- Logistic Regression Balanced Accuracy score: 0.702
- Random Forest Classifier Out-of-Bag balanced accuracy: 0.867

In Figures 6.1, 6.2 and 6.3 we present the feature importance ranking as well as SHAP values of the features for the OLS Linear Regression model, Logistic Regression, and Random Forest Classifier. Those plots below are a feature restriction on bear labelled market periods. In other words, out of the 729 rows that contains the weekly data from 2006 to 2019, we only consider bear rows (i.e. rows where the regime label is 1 or in which the probability of being bear is greater than 0.5) for our analysis which are 110 rows.

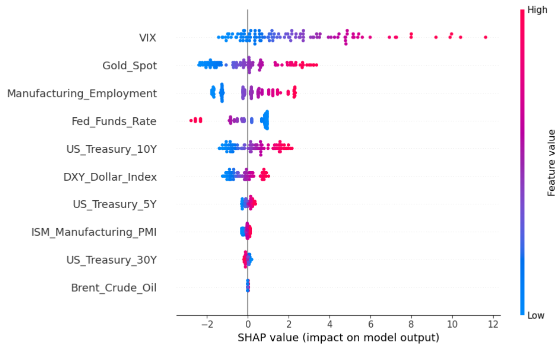


(a) SHAP analysis summary.

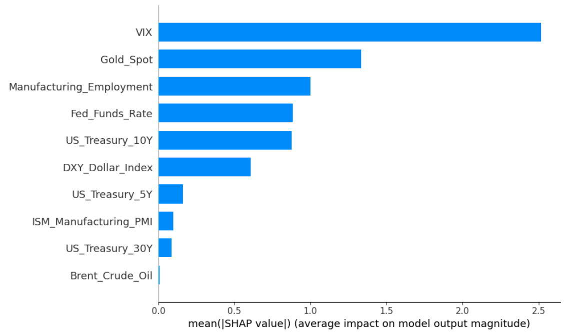


(b) Feature importance ranking.

Figure 6.1: Feature importance analysis for the OLS Linear Regression model fitted on the probabilistic target (Probability of being in a bear regime). The features are ranked based on their absolute coefficient values.



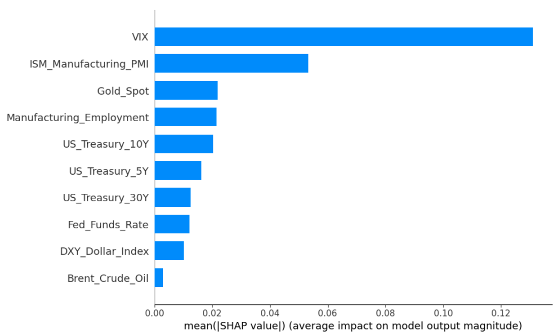
(a) SHAP analysis summary.



(b) Feature importance ranking.

Figure 6.2: Feature importance analysis for the Logistic Regression model fitted on the probabilistic target (Probability of being in a bear regime). The features are ranked based on their absolute coefficient values.

(a) SHAP analysis summary.



(b) Feature importance ranking.

Figure 6.3: Feature importance analysis for the Random Forest model fitted on the probabilistic target (Probability of being in a bear regime). The features are ranked based on their absolute coefficient values.

We observe that for all the supervised learning models – during bear periods – the most important macroeconomic indicator is the VIX index. This is consistent with the literature on financial markets, as the high VIX sensitivity indicates high market volatility and uncertainty which is typically associated with bear markets. The Linear Regression and Logistic Regression models also show that the gold spot is of high importance during bear periods. The high gold spot sensitivity also empirically coincides with bear markets as investors tend to rush to safe-haven assets such as gold during periods of market stress. Not only the feature rankings but also the SHAP analysis summary plots show this trend. We observe in the latter that elevated VIX and gold values (the red dots in the plots) push the model's output towards a higher probability of being in a bear regime (or towards a bear classification). Those high values also coincide with high SHAP value which shows a strong positive contribution of those features to the model's bear regime output.

On the other hand, we observe that the Fed Funds Rate captures high negative shap values when it is significantly elevated in the case of Linear Regression and Logistic Regression models. This suggests that higher Fed Funds Rate values push the model's output towards lower probability of being in a bear regime (or towards a 0 bull classification) which is also consistent with the literature as higher interest rates are typically associated with a strong economy and low market volatility.

Interestingly, the Random Forest model slightly deviates from the other two supervised learning models as it grants high feature importance to the Manufacturing PMI index. In particular, we obseve relatively high negative SHAP values for the above index when the index is elevated. Inversely, it gives high positive SHAP value when the index/feature is relatively low. This accurately describes how economic expansions and slowdowns match with bull and bear markets. However, the foward looking aspect of markets create laggs between market regimes and PMI index evolution. This may explain the fact that the other supervised learning models do not integrate PMI index as a major feature.

6.5 Summary

We observed that the SHAP analysis of the supervised learning models fitted on the sliced Wasserstein k-means clustering output shows a strong alignment between the detected bear regimes and the evolution of macroeconomic indicators during those periods. Key empirical observations of financial markets such as VIX and gold spot spikes coincide with bear labellings, especially in the lens of Linear Regression and Logistic Regression models. This suggests that the sliced Wasserstein k-means clustering algorithm labelling coincides with an empirical economic understanding of financial markets.

Chapter 7

Conclusion and Future Work

This chapter concludes the thesis and restates the key findings. It shows improvements as well as limitations of the proposed research and provides avenues for future work.

7.1 Summary

As we have seen in this thesis, the sliced Wasserstein k-means clustering algorithm is a well-suited method for detecting regimes in financial markets whether on synthetic or real-world data sets. Adding, sampling projections vectors using UnifOrtho elevates the total accuracy in high-dimensional settings ($d \geq 3$) as well as for low-dimensional settings ($0 < d \leq 2$) compared to the performance of the standard sliced Wasserstein k-means clustering algorithm with pre-specified projections in Luan and Hamp (2025) [11]. While, the time complexity of generating the UnifOrtho projections is higher ($\mathscr{O}(d^3)$) than maintaining pre-specified projections vectors, it is still very practical for increasingly high-dimensional data sets ($d \approx 100$) in which a significant number of projection vectors is needed to approximate the sliced Wasserstein distance.

When applying the sliced Wasserstein k-means clustering algorithm with UnifOrtho via look-ahead trading strategies on real-world data, we observe, as expected, very good performance indicating that the bear and bull regimes are well distinguished by the method. While this result shows promising results for observing contemperaneous bear and bull regimes, it is not sufficient to generate trading strategies. In fact, producing forecasting signals using the sliced Wasserstein k-means algorithm and implied probability of regime switch does not yield a highly profitable real-world trading strategy. For equity indices, in all regions tested (American, European and Asian equity clusters), the long-only benchmark – an inverse volatility weighted portfolio of indices – consistently outperforms the trading strategies based on the sliced Wasserstein k-means clustering algorithm. While, the trading strategies are able to flag many incoming bear periods notably through switch probabilities, they struggle to capture the upward momentum of financial markets in bullish/recovery periods as observed in the post-2020 test data.

On the interpretability side, the SHAP analysis of the sliced Wasserstein k-means clustering algorithm reveals an alignment between the detected bear regimes and the evolution of macroeconomic indicators during those periods. Key empirical observations of financial markets such as VIX and gold spot spikes coincide with bear labellings, especially in the lens of Linear Regression and Logistic Regression models.

7.2 Future Work

As we have seen previously, the UnifOrtho method approximating the sliced Wasserstein distance is a powerful tool to scale regime detection in high-dimensions. Yet it is relatively costly due to the QR decomposition required to generate the projection vectors. One way to improve the performance of the sliced Wasserstein k-means algorithm is to use Hadamard-Rademacher matrices to approximately sample UnifOrtho projections in $\mathcal{O}(d^2 \log d)$ time complexity. However, such complexity gain implies a trade-off in terms of clustering performance as the approximation of the sliced Wasserstein distance is less accurate than with pure UnifOrtho projections as constructed via the Haar distribution (Mezzadri (2007) [14]). For low dimensions to medium dimensions the exact UnifOrtho method in $\mathcal{O}(d^3)$ is still preferrable. However, for very high-dimensional data sets ($d \approx 1000$) the Hadamard-Rademacher method is a good alternative to reduce the computational cost of the sliced Wasserstein k-means clustering algorithm.

Another interesting avenue of research is to improve the implied probability trading strategies based on the sliced Wasserstein method. In experiment two we observed that the latter strategies struggle to capture upward momentum in bullish regimes. A possible improvement could be to combine regime-switch probability in combination with a regime-stay probability as a trading signal to avoid shorting during recovery or upward trend periods without necessarily going to cash.

.1 Appendix

.1.1 Sliced Wasserstein on Synthetic Data - Performance Tables

h_1	h_2	L	Max		Median		Max(MCCD)	
			A	B	A	B	A	B
10	2	2	0.5109	0.9715	0.5108	0.9715	0.5098	0.9711
		4	0.5858	0.9763	0.5141	0.9763	0.5858	0.9762
		9	0.5080	0.9442	0.5079	0.7253	0.5080	0.9442
		16	0.5091	0.9672	0.5087	0.7471	0.5090	0.9672
20	4	2	0.9744	0.9677	0.9741	0.9677	0.9740	0.9677
		4	0.9650	0.9947	0.9649	0.9946	0.9648	0.9946
		9	0.9734	0.9934	0.9733	0.9933	0.9733	0.9934
		16	0.9705	0.9954	0.9703	0.9953	0.9703	0.9954
30	6	2	0.9884	0.9862	0.9884	0.9860	0.9881	0.9853
		4	0.9864	0.9969	0.9863	0.9968	0.9863	0.9967
		9	0.9836	0.9939	0.9834	0.9939	0.9836	0.9939
		16	0.9858	0.9967	0.9858	0.9965	0.9858	0.9965
35	7	2	0.9895	0.9877	0.9895	0.9876	0.9895	0.9870
		4	0.9917	0.9962	0.9916	0.9961	0.9917	0.9962
		9	0.9883	0.9952	0.9881	0.9952	0.9883	0.9952
		16	0.9907	0.9957	0.9897	0.9955	0.9907	0.9957

Table 1: Synthetic Type A/B (K=2), dimension d=2: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{2,4,9,16\}$. Dark red ≈ 1, pale ≈ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			A	B	A	B	A	B
20	4	9	0.9857	0.9987	0.9856	0.9987	0.9856	0.9986
		16	0.9868	0.9980	0.9868	0.9980	0.9868	0.9980
		20	0.9811	0.9986	0.9810	0.9985	0.9810	0.9985
		40	0.9818	0.9979	0.9816	0.9978	0.9817	0.9978
30	6	9	0.9923	0.9981	0.9923	0.9981	0.9922	0.9979
		16	0.9927	0.9986	0.9926	0.9984	0.9925	0.9984
		20	0.9923	0.9978	0.9922	0.9978	0.9922	0.9977
		40	0.9935	0.9983	0.9934	0.9982	0.9935	0.9982
40	8	9	0.9965	0.9973	0.9964	0.9971	0.9965	0.9973
		16	0.9969	0.9980	0.9967	0.9980	0.9967	0.9980
		20	0.9960	0.9981	0.9959	0.9981	0.9960	0.9979
		40	0.9962	0.9981	0.9960	0.9981	0.9960	0.9981
50	10	9	0.9962	0.9967	0.9959	0.9967	0.9959	0.9964
		16	0.9964	0.9977	0.9964	0.9977	0.9964	0.9977
		20	0.9964	0.9973	0.9962	0.9973	0.9961	0.9973
		40	0.9964	0.9972	0.9963	0.9969	0.9964	0.9972
60	12	9	0.9959	0.9964	0.9956	0.9961	0.9956	0.9961
		16	0.9963	0.9974	0.9959	0.9974	0.9959	0.9971
		20	0.9959	0.9961	0.9956	0.9958	0.9956	0.9958
		40	0.9959	0.9968	0.9959	0.9965	0.9959	0.9965

Table 2: Synthetic Type A/B (K=2), dimension d=3: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 16, 20, 40\}$. Dark red $\approx$ 1, pale $\approx$ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			A	B	A	B	A	B
20	4	9	0.9866	0.9986	0.9866	0.9986	0.9866	0.9986
		16	0.9844	0.9985	0.9843	0.9985	0.9843	0.9985
		20	0.9867	0.9985	0.9866	0.9984	0.9865	0.9985
		40	0.9872	0.9988	0.9872	0.9988	0.9872	0.9988
30	6	9	0.9967	0.9981	0.9966	0.9980	0.9965	0.9979
		16	0.9931	0.9983	0.9930	0.9982	0.9930	0.9982
		20	0.9942	0.9982	0.9942	0.9981	0.9940	0.9981
		40	0.9943	0.9982	0.9943	0.9982	0.9943	0.9981
40	8	9	0.9954	0.9979	0.9953	0.9979	0.9954	0.9979
		16	0.9965	0.9981	0.9964	0.9979	0.9963	0.9979
		20	0.9969	0.9977	0.9968	0.9975	0.9967	0.9975
		40	0.9967	0.9981	0.9965	0.9979	0.9965	0.9979
50	10	9	0.9955	0.9973	0.9952	0.9973	0.9955	0.9973
		16	0.9948	0.9973	0.9948	0.9972	0.9948	0.9973
		20	0.9950	0.9973	0.9947	0.9971	0.9947	0.9973
		40	0.9948	0.9973	0.9946	0.9972	0.9946	0.9973
60	12	9	0.9951	0.9970	0.9951	0.9967	0.9948	0.9967
		16	0.9949	0.9972	0.9949	0.9972	0.9946	0.9969
		20	0.9951	0.9967	0.9951	0.9966	0.9951	0.9964
		40	0.9951	0.9966	0.9949	0.9963	0.9948	0.9966

Table 3: Synthetic Type A/B (K=2), dimension d=4: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 16, 20, 40\}$. Dark red ≈ 1, pale ≈ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			A	B	A	B	A	B
20	4	9	0.9985	0.9988	0.9985	0.9987	0.9985	0.9987
		20	0.9954	0.9990	0.9953	0.9990	0.9953	0.9990
		40	0.9955	0.9990	0.9955	0.9990	0.9955	0.9990
		100	0.9964	0.9991	0.9963	0.9991	0.9963	0.9991
30	6	9	0.9964	0.9986	0.9963	0.9985	0.9964	0.9986
		20	0.9976	0.9989	0.9975	0.9989	0.9976	0.9989
		40	0.9960	0.9989	0.9960	0.9987	0.9960	0.9987
		100	0.9974	0.9990	0.9973	0.9988	0.9973	0.9988
40	8	9	0.9978	0.9977	0.9976	0.9975	0.9978	0.9977
		20	0.9973	0.9984	0.9973	0.9984	0.9973	0.9984
		40	0.9968	0.9984	0.9968	0.9984	0.9966	0.9984
		100	0.9973	0.9984	0.9973	0.9984	0.9973	0.9984
50	10	9	0.9964	0.9982	0.9964	0.9982	0.9964	0.9980
		20	0.9972	0.9980	0.9969	0.9977	0.9972	0.9977
		40	0.9960	0.9973	0.9959	0.9972	0.9958	0.9970
		100	0.9969	0.9978	0.9966	0.9978	0.9966	0.9976
60	12	9	0.9950	0.9978	0.9948	0.9977	0.9950	0.9978
		20	0.9970	0.9974	0.9968	0.9973	0.9967	0.9971
		40	0.9966	0.9976	0.9963	0.9973	0.9963	0.9973
		100	0.9969	0.9981	0.9969	0.9981	0.9969	0.9981

Table 4: Synthetic Type A/B (K=3), dimension d=100: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 20, 40, 100\}$. Dark red $\approx$ 1, pale $\approx$ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			C	D	C	D	C	D
20	4	2	0.5328	0.8590	0.5304	0.5054	0.5304	0.8590
		4	0.9763	0.8423	0.5211	0.5172	0.9758	0.8423
		9	0.5310	0.8652	0.5271	0.5104	0.5271	0.8627
		16	0.9789	0.8518	0.5303	0.5088	0.9789	0.8499
30	6	2	0.9943	0.8800	0.9941	0.5146	0.9940	0.8800
		4	0.9936	0.8770	0.5481	0.5161	0.9929	0.8758
		9	0.8878	0.8746	0.5468	0.5061	0.8793	0.8746
		16	0.9927	0.5159	0.5429	0.5134	0.9922	0.5159
40	8	2	0.9906	0.8818	0.5506	0.5238	0.9897	0.8816
		4	0.9950	0.8819	0.9946	0.5270	0.9946	0.8814
		9	0.8723	0.8774	0.5449	0.5184	0.8723	0.8774
		16	0.9946	0.8812	0.7702	0.5268	0.9946	0.8812
50	10	2	0.8769	0.5277	0.5731	0.5166	0.8761	0.5035
		4	0.5756	0.5330	0.5736	0.5287	0.5739	0.5277
		9	0.9951	0.8951	0.5786	0.5254	0.9948	0.8939
		16	0.9942	0.8920	0.5765	0.5219	0.9942	0.8908
60	12	2	0.8909	0.9950	0.5711	0.5148	0.8909	0.9946
		4	0.9947	0.9027	0.7822	0.5324	0.9944	0.9027
		9	0.9944	0.8918	0.5800	0.6979	0.9941	0.8901
		16	0.9964	0.9032	0.9958	0.5240	0.9955	0.9032

Table 5: Synthetic Type C/D (K=3), dimension d=2: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{2, 4, 9, 16\}$. Dark red ≈ 1, pale ≈ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			C	D	C	D	C	D
20	4	9	0.9825	0.8584	0.5422	0.5120	0.9825	0.8579
		16	0.9852	0.5174	0.5404	0.5144	0.9852	0.5148
		20	0.9860	0.5202	0.9858	0.5170	0.9858	0.5186
		40	0.9849	0.8612	0.7620	0.5137	0.9849	0.8608
30	6	9	0.9930	0.8730	0.9928	0.5063	0.9930	0.8713
		16	0.9930	0.5150	0.5843	0.5121	0.9930	0.5097
		20	0.9931	0.8745	0.5758	0.5056	0.9931	0.8745
		40	0.9931	0.8763	0.5786	0.5075	0.9930	0.8752
40	8	9	0.9939	0.8832	0.5690	0.5032	0.9939	0.8832
		16	0.9943	0.8832	0.7807	0.5039	0.9943	0.8830
		20	0.9948	0.8781	0.9944	0.5057	0.9944	0.8781
		40	0.9941	0.8846	0.5653	0.5066	0.9939	0.8819
50	10	9	0.9954	0.8779	0.9949	0.5050	0.9954	0.8776
		16	0.5649	0.8798	0.5617	0.5072	0.5603	0.8782
		20	0.9957	0.8772	0.7788	0.5103	0.9952	0.8771
		40	0.9955	0.8783	0.5552	0.5092	0.9955	0.8783
60	12	9	0.9952	0.5169	0.5590	0.5066	0.9952	0.5061
		16	0.9959	0.8767	0.9955	0.5126	0.9959	0.8767
		20	0.9952	0.8789	0.9949	0.5156	0.9949	0.8789
		40	0.9955	0.8746	0.5660	0.5111	0.9952	0.8746

Table 6: Synthetic Type C/D (K=3), dimension d=3: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 16, 20, 40\}$. Dark red ≈ 1, pale ≈ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			C	D	C	D	C	D
20	4	9	0.9895	0.8652	0.5501	0.7030	0.9895	0.8644
		16	0.9873	0.8660	0.7748	0.5388	0.9872	0.8648
		20	0.9887	0.8682	0.7770	0.5446	0.9884	0.8682
		40	0.5707	0.5465	0.5677	0.5437	0.5669	0.5462
30	6	9	0.9935	0.8738	0.5788	0.5284	0.9935	0.8738
		16	0.9935	0.8735	0.5737	0.5258	0.9935	0.8735
		20	0.9942	0.8720	0.5778	0.5240	0.9942	0.8720
		40	0.9942	0.8718	0.7824	0.5265	0.9940	0.8718
40	8	9	0.9950	0.8732	0.5569	0.5241	0.9949	0.8732
		16	0.9952	0.8717	0.5637	0.5375	0.9952	0.8712
		20	0.9963	0.8723	0.5555	0.5399	0.9958	0.8723
		40	0.9955	0.8719	0.5649	0.5419	0.9952	0.8719
50	10	9	0.9946	0.8773	0.5689	0.7008	0.9946	0.8771
		16	0.9947	0.5239	0.5556	0.5223	0.9937	0.5211
		20	0.9950	0.8741	0.5553	0.5247	0.9950	0.8738
		40	0.9950	0.8732	0.5602	0.5276	0.9937	0.8724
60	12	9	0.9944	0.8733	0.5767	0.5405	0.9940	0.8733
		16	0.9936	0.8757	0.5622	0.5437	0.9929	0.8757
		20	0.9948	0.8764	0.5591	0.5447	0.9948	0.8746
		40	0.9955	0.8744	0.5688	0.5451	0.9955	0.8744

Table 7: Synthetic Type C/D (K=3), dimension d=4: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 16, 20, 40\}$. Dark red ≈ 1, pale ≈ 0.5.

h_1	h_2	L	Max		Median		Max(MCCD)	
			C	D	C	D	C	D
20	4	9	0.9966	0.8734	0.5821	0.5558	0.9965	0.8728
		20	0.9954	0.8775	0.5722	0.5575	0.9949	0.8755
		40	0.9981	0.8754	0.5947	0.5510	0.9981	0.8751
		100	0.9960	0.8745	0.5909	0.5508	0.9959	0.8745
30	6	9	0.9972	0.8785	0.5749	0.5586	0.9972	0.8774
		20	0.9973	0.8803	0.6002	0.5578	0.9963	0.8797
		40	0.9976	0.8841	0.7981	0.5526	0.9966	0.8836
		100	0.9967	0.8825	0.6028	0.5536	0.9962	0.8780
40	8	9	0.9941	0.8747	0.6009	0.5538	0.9933	0.8741
		20	0.9963	0.8791	0.6280	0.5560	0.9963	0.8791
		40	0.9973	0.8827	0.6226	0.5577	0.9973	0.8780
		100	0.9968	0.8806	0.6208	0.5570	0.9968	0.8779
50	10	9	0.9939	0.8762	0.6139	0.5735	0.9939	0.8754
		20	0.9953	0.8836	0.6050	0.5666	0.9942	0.8836
		40	0.9965	0.8796	0.6094	0.5589	0.9965	0.8756
		100	0.9957	0.8832	0.6259	0.5624	0.9957	0.8761
60	12	9	0.9963	0.8775	0.9947	0.5695	0.7746	0.8758
		20	0.9947	0.8881	0.6101	0.5588	0.9939	0.8864
		40	0.9938	0.8823	0.6282	0.5538	0.9935	0.8821
		100	0.9950	0.8804	0.6151	0.5603	0.9942	0.8781

Table 8: Synthetic Type C/D (K=3), dimension d=100: total accuracy of the SW k-means clustering with UnifOrtho projections across window size h_1, step h_2 and number of projections $L \in \{9, 20, 40, 100\}$. Dark red $\approx$ 1, pale $\approx$ 0.5.

Look-ahead regime strategy

Algorithm 2: LOOKAHEADSTRATUNIFORTHO — Regime Strategy with Look-Ahead Bias

Input: Initial capital C_0,;
number of simulations N_S,;
price matrix $S \in \mathbb{R}^{T \times d}$,;
number of projections L,;
trimming parameters h_1, h_2,;
window size w,;
number of clusters K=2,;
risk metric m,;
weighting scheme
Output: Portfolio value path V,;
signal vector s,;
cumulative PnL

$\varepsilon \leftarrow 10^{-6}$;
if *weighting* = "equal" **then**
    $R \leftarrow \mathrm{PctReturns}(S)$; ▷ drop first (NaN) row
    $\theta \leftarrow \mathbf{1}_d / d$; ▷ equal weights
    $r^{(\mathrm{pf})} \leftarrow R\,\theta^{\top}$; ▷ row-wise sum
else
    $R \leftarrow \mathrm{PctReturns}(S)$;
    $\sigma \leftarrow \mathrm{RollingStd}(R, w)$, keep rows $\geq w-1$;
    $u \leftarrow \mathbf{1}/(\sigma + \varepsilon)$; ▷ inverse vol, elementwise
    $\theta_t \leftarrow u_t / \sum_j u_{t,j}$; ▷ normalize each row to sum 1
    $r_t^{(\mathrm{pf})} \leftarrow \sum_j R_{t,j}\,\theta_{t,j}$; ▷ aligned from row $w-1$
$\hat{F}$, $\boldsymbol{\mu}$, $\boldsymbol{\lambda} \leftarrow$ MAXMCCD_UNIFORTHO$(N_S, S, K, L, \varepsilon, h_1, h_2, m)$;
for *each t* **do**
    if $\lambda_t = 1$ **then**
        $s_t \leftarrow +1$; ▷ Bullish ⇒ Long
    else
        $s_t \leftarrow 0$; ▷ Bearish ⇒ Flat
▷ Step 4 -- Apply signal & accounting
$r_t^{(\mathrm{strat})} \leftarrow s_{w-h_1+t} \cdot r_t^{(\mathrm{pf})}$; ▷ contemporaneous → look-ahead bias
$\mathrm{cum}_t \leftarrow \prod_{k \leq t} \left(1 + r_k^{(\mathrm{strat})}\right)$;
$V_t \leftarrow C_0 \cdot \mathrm{cum}_t$;
$\mathrm{PnL}_t \leftarrow V_t - C_0$;
return V, s, PnL

SW k-means Long/Short trading strategy

Algorithm 3: SW_K-MEANS_LONG/SHORT — Binary Regime Strategy

Input: Initial capital C_0,;
number of simulations N_S,;
price matrix $S \in \mathbb{R}^{T \times d}$,;
trimming parameters h_1, h_2,;
window size w,;
number of clusters K,;
risk metric m,;
majority look-back ℓ

Output: Portfolio value path V

$R \leftarrow \text{PctReturns}(S)$; ▷ $R_t = S_t / S_{t-1} - 1$
$\theta \leftarrow \mathbf{1}_d / \sigma_d$; ▷ `inverse vol weighted returns`
$r^{(\text{pf})} \leftarrow R\theta$; ▷ `portfolio return series`
$V \leftarrow [C_0]$;
$n \leftarrow \lfloor T/w \rfloor$;
for $i = 0$ **to** $n-2$ **do**
  $s \leftarrow i \cdot w, \quad e \leftarrow (i+1) \cdot w$;
  $W \leftarrow S[s:e]$; ▷ `current analysis window`
  $\hat{F},\ \boldsymbol{\mu},\ \boldsymbol{\lambda} \leftarrow$ MAXMCCD_UNIFORTHO$(N_S, W, K, L, \varepsilon, h_1, h_2, m)$;
  if $\ell > |\boldsymbol{\lambda}|$ **then**
    $\tilde{\boldsymbol{\lambda}} \leftarrow \boldsymbol{\lambda}$;
  else
    $\tilde{\boldsymbol{\lambda}} \leftarrow \boldsymbol{\lambda}_{[-\ell:]}$;
  for $k = 0, 1, \ldots, |\tilde{\boldsymbol{\lambda}}| - 1$ **do**
    $w_k \leftarrow \exp\left(-\frac{\ln 2}{\tau}\left(|\tilde{\boldsymbol{\lambda}}| - 1 - k\right)\right)$;
  $\hat{k} \leftarrow \arg\max_k \sum_{i=0}^{|\tilde{\boldsymbol{\lambda}}|-1} w_i \mathbf{1}\left[\tilde{\lambda}_i = k\right]$;
  if $\hat{k} = 1$ **then**
    $\sigma \leftarrow +1$; ▷ `Bullish ⇒ Long`
  else
    $\sigma \leftarrow -1$; ▷ `Bearish ⇒ Short`
  for *each* $r_t \in r^{(\text{pf})}[e : e+w]$ **do**
    V.append$\Big(V[-1] \times \big(1 + \sigma r_t\big)\Big)$;
return V

Implied Probability SW k-means trading strategy

Algorithm 4: LONG_STRAT_IMPLIED — Implied-Probability Regime Strategy

Input: Initial capital C_0; simulations N_S; price matrix $S \in \mathbb{R}^{T\times d}$; trimming h_1, h_2; window w; clusters K; metric m; signal type $\tau \in \{\texttt{continuous}, \texttt{hysteresis}, \texttt{conviction}\}$; entry/hold thresholds α_e, α_h; look-back ℓ; gradient flag/weight (g, ω_g)

Output: Portfolio value path V; signal history Σ

$R \leftarrow \mathrm{PctReturns}(S)$; $\theta \leftarrow \mathbf{1}_d \cdot 1/\sigma_d$; $r^{(\mathrm{pf})} \leftarrow R\theta$;
$V \leftarrow [C_0]$; $\Sigma \leftarrow []$; $n \leftarrow \lfloor T/w \rfloor$;
for $i = 0$ **to** $n-2$ **do**
 $s \leftarrow i \cdot w$; $e \leftarrow (i+1)\cdot w$; $W \leftarrow S[s:e]$;
 $\hat{F}, \boldsymbol{\mu}, \boldsymbol{\lambda} \leftarrow$ MAXMCCD_UNIFORTHO$(N_S, W, K, L, \varepsilon, h_1, h_2, m)$;
 $\mathbf{P}, p_{\mathrm{switch}}, \mathbf{T}, \boldsymbol{\pi} \leftarrow$ COMPUTEIMPLIEDPROBA$(\hat{F}, \boldsymbol{\mu}, \boldsymbol{\lambda}, \ell, \mathrm{g}, \omega_g)$;
 if $\ell > |\boldsymbol{\lambda}|$ **then**
  $\tilde{\boldsymbol{\lambda}} \leftarrow \boldsymbol{\lambda}$;
 else
  $\tilde{\boldsymbol{\lambda}} \leftarrow \boldsymbol{\lambda}_{[-\ell:]}$;
 for $k = 0, 1, \dots, |\tilde{\boldsymbol{\lambda}}| - 1$ **do**
  $w_k \leftarrow \exp\left(-\dfrac{\ln 2}{\tau}\,(|\tilde{\boldsymbol{\lambda}}| - 1 - k)\right)$;
 $\hat{k} \leftarrow \arg\max_k \sum_{i=0}^{|\tilde{\boldsymbol{\lambda}}|-1} w_i \mathbf{1}\left[\tilde{\lambda}_i = k\right]$;
 if $\tau = \texttt{continuous}$ **then**
  $\sigma \leftarrow \pi_{\mathrm{bull}} - \pi_{\mathrm{bear}}$;
  if $|\sigma| < \delta$ **then** $\sigma \leftarrow 0$ ($\triangleright$ dead zone);
 else if $\tau = \texttt{hysteresis}$ **then**
  $\sigma_{\mathrm{prev}} \leftarrow \Sigma[-1]$ **or** 0;
  if $\hat{k} = \mathit{Bull}$ **then**
   if $p_{switch} \geq \alpha_e$ ***and*** $\sigma_{prev} \geq 0$ **then** $\sigma \leftarrow -1$;
   else if $p_{switch} < \alpha_h$ ***and*** $\sigma_{prev} < 0$ **then** $\sigma \leftarrow +1$;
   else $\sigma \leftarrow 0$;
  else $\triangleright\ \hat{k} = \texttt{Bear}$
   if $p_{switch} \geq \alpha_e$ ***and*** $\sigma_{prev} \leq 0$ **then** $\sigma \leftarrow +1$;
   else if $p_{switch} < \alpha_h$ ***and*** $\sigma_{prev} > 0$ **then** $\sigma \leftarrow -1$;
   else $\sigma \leftarrow 0$;
 else if $\tau = \texttt{conviction}$ **then**
  $d_k \leftarrow \begin{cases} +1 & \hat{k} = \mathrm{Bull} \\ -1 & \hat{k} = \mathrm{Bear} \end{cases}$;
  $c \leftarrow 1 - 1.5\,p_{\mathrm{switch}}$; $\triangleright$ `conviction decays with switch risk`
  $\sigma \leftarrow d_k \cdot c$;
 $\Sigma.\mathrm{append}(\sigma)$;
 for *each* $r_t \in r^{(\mathrm{pf})}[e : e+w]$ **do**
  $V.\mathrm{append}\big(V[-1] \times (1 + \sigma r_t)\big)$;
return V, Σ

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